TEMPLATE

1.1 | Key Project Information & project Design Document (PDD)

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VERSION **v. 1.3**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.3

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KEY PROJECT INFORMATION

GS ID of Project	GS11138
Title of Project	National Biodigester Programme Phase IV
Time of First Submission Date	03/04/2021
Date of Design Certification	/
Version number of the PDD	4.1
Completion date of version	5/04/2024
Project Developer	National Biodigester Programme
Project Representative	National Biodigester Programme, Cambodia
Project Participants and any communities involved	N/A
Host Country (ies)	Cambodia
Activity Requirements applied	☒ Community Service Activity ☐ Renewable Energy ☐ Land-Use and Forests Activity Requirements/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	
Methodology (ies) applied and version number	Methodology for Animal Manure Management and Biogas use for Thermal Energy Generation, version 1.1 AMS-I.F.; Renewable electricity generation for captive use and mini-grid, version 5.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

SUSTAINABLE DEVELOPMENT GOALS TARGETED	SDG IMPACT (DEFINED IN B.6)	ESTIMATED ANNUAL AVERAGE	UNITS OR PRODUCTS
13 Climate Action (mandatory)	The total amount of Emission Reduction	21,717	tCO_2e/yr GS VERs
1 No Poverty	Financial savings from avoided biomass purchase	85,792	USD/yr
2 Zero hunger	Total Area on which bio-slurry is applied	812	ha/yr
7 Affordable and Clean Energy	Household access to clean energy with the biodigesters sold under the project activity	667	#HH
7 Affordable and Clean Energy	Electricity consumption by users produced by the project Renewable energy	3,015	MWh/yr
8 Decent Work and Economic Growth	Number of Jobs Created by the project activity	80	#employment
15 Life on Land	Forest Area Saved	6.702	ha/yr

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

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Purpose of the project

The overall objective of the National Biodigester Programme is the dissemination of domestic biodigesters as an indigenous, sustainable energy source through the development of a commercial, market oriented, biodigester sector in at least seventeen selected provinces of Cambodia. The specific objectives of the National Biodigester Programme contributing to its overall objective are:

1. To increase the number of family sized, quality biodigesters with the total 5,000 biodigesters in the period 2021-2026 in selected provinces. The programme will also promote the installation of biogas or biogas-diesel generator at medium and large size anaerobic lagoon digesters. It is aiming to install about 120 of those generators.

2. To ensure the continued operation of all biodigesters installed under the biodigester programme;
3. To maximise the benefits of the operated biodigesters, in particular the optimum use of digester effluent;
4. Technical and promotional capacity development of the stakeholders within the NBP for further wide scale deployment of biodigester technology in Cambodia. This objective will particularly focus on the development of a capable and viable private sector responsible for marketing, construction and after-sales service of biodigesters.

NBP supports three technologies, two masonry fixed dome models (Farmer's friend and the NBP S1) and a prefabricated composite digester (a Vietnamese digester design). Biodigester Construction Agents (BCAs) are responsible for constructing or installing biodigesters and operate under a franchise contract with NBP. Around 77 BCAs are currently active which operate in a semi-commercialized manner and employ 854 masons who are certified for biodigester construction. Another important benefit therefore is employment generation in rural areas.

Next to this, NBP has installed a few number of medium and large size anaerobic lagoon digesters at animal farms as part of their pilot projects with the support from other project/programme including Cambodia Climate Change Alliances (CCCA) and Building Adaptive Capacity through the Scaling-Up of Renewable Energy Technologies in Rural Cambodia (S-RET) Project.

Technology to be employed/implemented
Under its phase IV, NBP continue: (1) building household biodigester in a similar way as its previous phases and (2) working with private sector to promote the installation of biodigester at animal farm (small, medium and large-scale farm) with generator and/or flaring system for better animal waste management and reducing electricity bill at the farm.
Biodigesters will hygienically treat animal and human waste in a biodigester to prevent the spread diseases from animal to animal, animal to human and human to animal and produce a clean renewable cooking fuel, biogas, whereas the treated waste is to be used as a potent and safe organic fertilizer.

Project Scenario

The project will: (1) replace traditional cooking devices using fossil fuels or non-renewable biomass with biogas stove, improve the animal waste management and produce organic fertilizer for soil quality improvement at household level; and (2) save electricity bills and fossil fuel and improving AWMS at animal farm.
Therefore, the project will provide economic, social and environmental benefits to users at both households and animal farm level. It will also contribute to the reduction of greenhouse gas emissions.

The scenario existing prior to the project activity
At household level, before the onset of the project activities, most households with the technical potential for a biodigester rely primarily on wood for cooking both causing substantial exposure to hazardous household air pollution (with related health hazards) and contribution to deforestation. A substantial part of the fuel wood is collected, which is both drudgery and significant time expenditure for especially women. Purchased wood on the other hand is a burden on the limited household's revenues. In addition, unhygienic animal waste management practices and the lack of access to basic sanitation result in pollution, foul odor, methane emissions and a relatively high prevalence of hygiene related diseases, such as diarrhea.

At animal farms in Cambodia, the manure management is still limited. Although Ministry of Agriculture Forestry and fisheries (MAFF) issues Prakas No549 on 12 Dec 2018 by mandating the treatment of animal waste (covering lagoon) at the farm, the application of biogas generating from the treatment system is still lacking. It is observed that some farms just cover the lagoon while the generated biogas is released into the atmosphere without flaring or using as fuel for generating electricity or thermal. Also, animal farms consumed electricity from the national grid for running the evaporative cooling system and lighting.

A.1.1. Eligibility of the project under Gold Standard

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The eligibility criteria as per GS4GG Principles & Requirements, v1.2 are explained below:
(a) Types of projects:
As per para 4.1.3, of GS4GG principles & Requirements v1.2, "A project type is automatically eligible for Gold Standard Certification if there are Gold Standard

approved Activity Requirements and/or Impact Quantification Methodologies associated with it or it's referenced in the Gold Standard Product Requirements. These are published to the Gold Standard website and shall be followed where provided for a given project type"

The project will include the distribution of Household and animal farms biogas units (Waste Management), so it is automatically eligible under the project type categories as defined in the GS Activity requirements: "Waste Management and handling". Besides, the project uses Gold Standard Methodology "METHODOLOGY FOR ANIMAL MANURE MANAGEMENT AND BIOGAS USE FOR THERMAL ENERGY GENERATION" version 1.0 and CDM Methodology "AMS-I.F. Renewable electricity generation for captive use and mini-grid" version 05.0
Hence the project type is eligible under GS4GG.
(b) Location of project:
The location of the Project is Cambodia. This is eligible as per the GS4GG eligibility requirements which specify that the projects can be in any part of world.

(c) Project area, Project Boundary & Scale:
The Project Area is Cambodia nationwide but starting from 17 provinces first as defined in A.2.
The Boundary is defined in B.3.
The project scale is small scale as defined below in section A.4.

The Project is only included under the Gold Standard and no dual certification will take place. The proposed project is the only active domestic biogas carbon project in the Project Area
(d) Host Country Requirements:
No legal, environmental, ecological and social regulations in Cambodia prevent implementation of biodigesters in the rural households and animal farms. Hence, the project follows the Host Country's legal, environmental, ecological and social regulations.
(e) Contact Details
The contact details of the project participants are provided Appendix 2 of this document.
(f) Legal Ownership:

PD clearly communicated to the stakeholders that PP will claim ownership rights at the stakeholder consultation and no question raised related to that no comment received. In addition, the PP explains and sign the form 03 and form 12 that the PP retains the rights of ownership of the GHG reductions to end-users before/at the installation of the biodigesters as defined below in section A.1.2.

(g) Other Rights:

Except the Social benefits and VERs, all other legal rights of the project devices including the use of biogas slurry are with the beneficiaries. PD will inform GS of any disputes.

(h) ODA Declaration:

No ODA fund is involved. The Project representative has provided ODA declaration conforming the same.

Hence, the project satisfies all the Gold Standard General Eligibility criteria.

The eligibility criteria as per section 3 of GS4GG Community Services Activity Requirements, V1.2 are explained below:

1) Types of project:

   The project covers type a) Renewable energy with the hybrid or pure biogas generators in animal farm level, type b) End-use energy efficiency with the integrated biogas stoves which replaced traditional cooking devices using fossil fuels or non-renewable biomass and type c) Waste management and handling in both levels by improving animal waste management and produce organic fertilizer for soil quality improvement in households and improved AWMS in the animal farm level.

2) Project area, boundary and scale:

   The project boundary will be limited to animal farms and households in the geographical boundary of Cambodia.

   Project is small Scale as described below in section A.4

3) Legal Ownership:

   The biodigester owners in household levels and animal farm levels are transferring rights of carbon credits ownership to NBP with the form 03 and form12 which described below section A.1.2. The transfer of Product ownership also discussed during local stakeholder consultation.

The eligibility criteria as per section 2 of GS4GG GHG Emissions Reduction & Sequestration Product Requirements, V2.1 are explained below:

1) Unless otherwise stated elsewhere in the Principles & Requirements, Projects involving a mix of eligible and ineligible components can only claim credits for the Emission Reductions and/or sequestration associated with the eligible component of the project.

This project will be only claim Gold Standard Verified Emission Reductions (GSVERs), emission reductions associated with the eligible component of the project.

2) Bundled Projects: Projects submitted together for certification within a bundle, each Project that is part of bundle shall individually conform to all GS4GG Requirements.

Project is not a bundled project, this project is a standalone project.

3) Voluntary project activities (VPAs): A group of Projects submitted together for Gold Standard Design Certification within a Programme of Activities.

This project is not consisting any VPAs, however conform all the requirements of Community Services Activity Requirements listed above

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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Households that invest in a biodigester sign a sales contract, called **form 03**. In that form, a clause is included on the VER rights transfer from the households to NBP. This clause, No 19 states: "Transfers all rights, credits, entitlements, benefits or allowances arising from or in connection with any greenhouse gas emissions reductions arising from the operation of the biodigester (Emission Reductions), and agrees to take all necessary action required to ensure the transfer of those Emission Reductions to the National Biodigester Programme".

Animal farms' owners who invested in covered lagoon biodigester and installed biogas generator agreed to transfer the carbon credit right to NBP and it is up to NBP to manage those resources to support the development of biodigester sector as stated in **form 12.**

The transfer of Product ownership discussed during local stakeholder consultation for household level and animal farm level projects and informed the end users that they cannot claim emission reductions from the project.

A.2 Location of project

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NBP is currently active in the following provinces:

Table 2: GPS coordinates of the provincial capital in the current NBP provinces

#	Province	Latitude (xx° xx' xx" N)	Longitude (xx° xx' xx" E)
1	Kampong Cham	11° 59' 00" N	105° 27' 00" E
2	Kandal	11° 78' 30" N	104° 81' 70" E
3	Svay Reang	11° 05' 00" N	105° 48' 00" E
4	Takeo	10° 59' 00" N	104° 47' 00" E
5	Kampong Speu	11° 27' 00" N	104° 30' 00" E
6	Kampong Chhnang	12° 00' 00" N	104° 30' 00" E
7	Kampot	10° 36' 00" N	104° 10' 00" E
8	Kampong Som	10°38’00” N	103°30’00” E
9	Kep	10°29’00” N	104°18’00” E
10	Prey Veng	11°29' 00" N	105°19' 00" E
11	Siem Reap	13°21’44” N	103°51’35”E
12	Pursat	12°32’00” N	103°55’00” E
13	Battambang	13°06’00” N	103°12’00” E
14	Kampong Thom	12°42’00” N	104°53’00” E
15	Tbong khmum	11°54’34”N	105°38’49”E
16	Phnom Penh	11°33’43” N	104°53’18” E
17	Banteay Meanchey	13°45’00” N	103°00’00” E

NBP has the long term aim to meet the biogas potential in the whole country depending on the budget available to enlarge the geographical scope of the Program. All the households with the technical potential (>15kg of manure at their disposal daily) within the project area are targeted, irrespective of district, commune or village. Likewise, for covered lagoon biodigester at farm, the size of biodigester is determined by the biodigester technician under the coordination with NBP, based on the number of animal per round.

As the objective of the programme is to establish a commercially viable market for domestic biogas, the biodigester dissemination will follow market demand. Consequently, location details will only be available after households and biogas construction enterprises have entered into a contractual agreement.

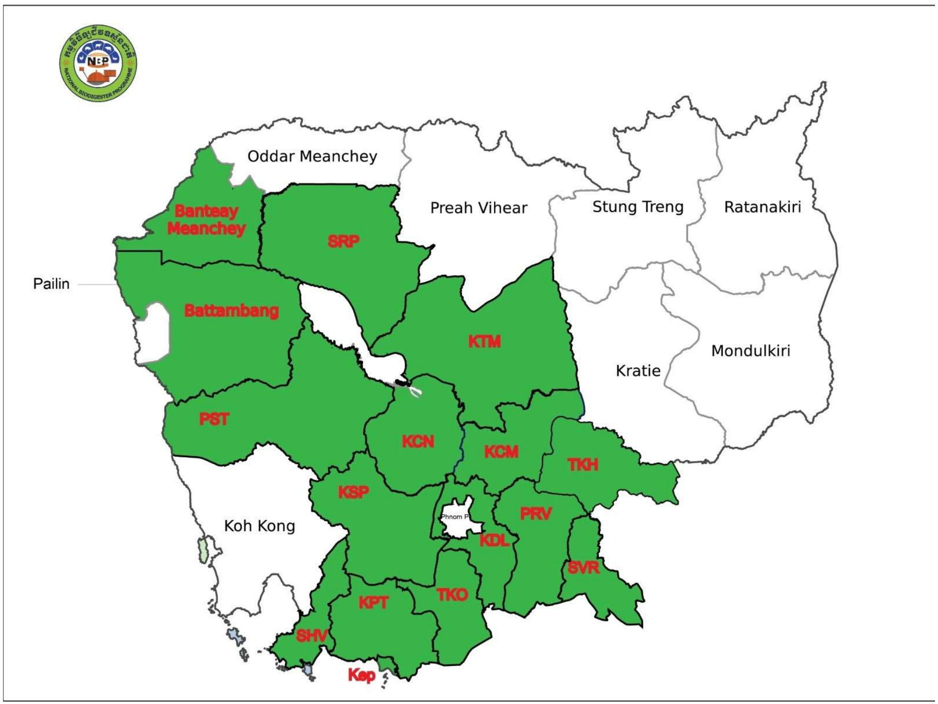

Figure 1 Provinces in which NBP is active

A.3 Technologies and/or measures

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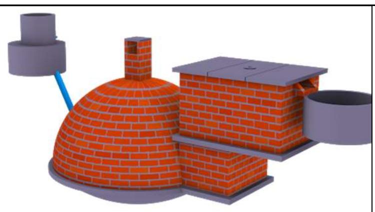
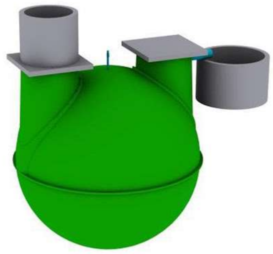
For domestic biodigester, NBP installs, through franchised Biodigester Company Agencies (BCA), biodigesters. The main type of digester that NBP installs is the Farmer's Friend digester, with possibility to construct up to 50 m^3 a scaled down version of this model, the S1 in the size of 2 to 3 m^3 and composite digesters. The volumetric biogas production is around 1 m^3 biogas per 3 m^3.

Table 3 Specifications of Household digesters models

Type	Farmer's Friend	NBP S1	Composite
Size (m3)	4-15 with option to 50m^3	2-3	4m^3, 7m^3, 9m^3
Feeding (kg/day)	30-40 (4m3) to 100-150 (15 m3)	10-20 (2m3) 20-30 (3m3)	30-40 (4m3), 60-70 (7m3), 80-90 (9m3)

Digester drawing			

All different type of household digester have an expected lifespan of over 20 years. The programme is otherwise biodigester technology agnostic, provided that technologies meet the following requirements:

- Durability: Biodigesters are expected to last for over 10 years
- Gas storage: The digester should be able to store at least 50% of daily gas production[1]
- Warranty: A minimum warranty of 1 year shall be offered

The list of facilities, system and equipment that will be installed include:

- Inlet (for mixing the manure with water)
- Biodigester and integrated gasholder
- Compensation tank
- Slurry pit (to store overflown slurry temporarily before it is scooped to the compost hut)
- Compost hut (optional but strongly encouraged)
- Toilet (optional)
- Gas piping and water trap (water trap is necessarily to remove condense water from the biogas)
- A biogas stove (stoves disseminated via NBP are recommended but not mandatory)
- Pressure gauge (gas pressure in the plant is a proxy for gas availability)
- Biogas lamp / biogas rice cooker (both optional)

[1] Plant size calculation sheet for Farmer's Friend and NBP S1.

NBP is a market-based programme and the actual number of digester installed depends on the market conditions. NBP however forecasts the uptake of household biodigesters in this CP.

The other technology is for the animal farms where the covered lagoon small, medium or large scale biodigesters (SMLSB) installation occurs to manage animal waste and Farm biodigester, NBP is cooperating with private sector to promote and install covered lagoon biodigester at animal farm and has installed a number of medium and large size anaerobic lagoon digesters.

Covered lagoon biodigester been designed and piloted with the animal farms especially pig farm. The generated biogas is used to run electricity generator. The generated electricity is then used complementally to electricity grid to run water pump for cleaning pigs stable and evaporative colling system.

Type	**Covered Lagoon**
Size (m3)	50-100 m3 depending on scale
Feeding (kg/day)	500-10000 kg/day
Digester drawing	

The main components of the covered lagoon digester system should include 6 units and optional unit as described below, please refer to Figure 4 Covered Lagoon Design:

1) Collecting tank, used to collect all wastewater discharged from the farm.
2) Digester influent feeding into the covered lagoon digester.
3) Covered lagoon digester, used for digesting wastewater under anaerobic condition.
4) Digestate storage lagoon, used for storing digestate pushed out of the covered lagoon digester.
5) Biogas cleaning system, removing water and hydrogen sulfide.
6) Biogas utilization system, under which biogas is drawn to produce electricity and

heat, or to use biogas as a cooking fuel

7) Flare system (optional) for flaring the excessive biogas.

A.4 Scale of the project

The project is a small-scale project. The estimated technical potential for biodigesters in Cambodia, as a small-scale project is limited to 45 MWth, or around 13,211 households units where the average energy output is 0.749kW/m^3 according to calculations[2]. The other threshold by the requirements is 60,000 tCO_2eq per annum in any year of the crediting period, where it is expected to have 3,408 unit in first 5 year and highest ex-ante ER 45,616 in the year 5, for both threshold project is below so the project is small scale.

A.5 Funding sources of project

>> NBP received a mix of funding sources, including VER income, Polastern. The funding breakdown is shown here below: "The total expenditure, including subsidy by the end of Jan-December 2022 was US$ 568,364 of which 87% from VER fund (carbon finance), 12% from Polarstern fund and 1% from CCC"[3]. **While, Polarstern is** a German electricity provider company (https://www.polarstern-energie.de/), has cooperated with NBP by providing grant for subsidy to farmers and loan-based fund to NBP and all the carbon credits generated from this fund are belong to NBP as stated in its cooperate agreement[4]; CCC is an NGO based in the Republic of Korean has contracted with NBP to build biodigester for some targeted farmers as detail in the contract agreement[5]. Furthermore, ODA declaration forms signed by NBP, is provided to confirm that such funding does not result in a diversion of official development assistance.

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[2] ER spreadsheet, tab Project Scale.
[3] NBPC 2022 Annual Report
[4] See cooperate agreement between NBP and Polarstern
[5] See contract agreement between NBP and CCC

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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Methodology For Animal Manure Management and Biogas Use For Thermal Energy Generation V1.1[6] is employed for AWMS and thermal applications at the household level. Additionally, it is utilized for AWMS applications within animal farms. Moreover, considering the electricity application at the animal farm level, the methodology AMS-I.F.: Renewable electricity generation for captive use and mini-grid V5.0[7]
CDM Tool 3: Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion V03.0[8] is utilized for calculating grid electricity replacement.

B.2. Applicability of methodology (ies)

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According to the Baseline and Monitoring Methodology Biodigester v1.1; in the baseline scenario which determined by survey and defined below in section B.4 shows the fraction of in absence of the project animal manure was left to decay anaerobically and the methodology only applied to the part is left to decay.
Also stated, where annual emission reduction for methane recovery component is higher than five tonnes of CO_2eq per biodigester the AWMS method 2 shall be applied. In the project average Ex-ante emission reductions from AWMS applications, are above five tCO2e, regarding that, the AWMS method 2 - IPCC Tier 2 approach is used.

Applicability criteria for GS Methodology which applied for both household level and animal farm level, AWMS method 2 approach listed below in the table:

Eligibility criteria	Assessment

[6] https://globalgoals.goldstandard.org/433-ee-ics-methodology-for-animal-waste-management-and-biogas-application/
[7] https://cdm.unfccc.int/UserManagement/FileStorage/I1P3XEBCJ0625DUALGSNO79R8FWY4V
[8] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf

The livestock population in the farm is managed fully or partly under confined conditions	At households level, the information of animal raising and its number can be determined by monitoring survey, asking to household owner.
	All the records of buy – sold pigs included if any deaths recorded by the farms. Fattening animal farms with mother pigs, store all information about the newborns.
Manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries);	No discharge into natural water resources in the project scenario[9]. Also, the substance (bio-slurry) after the treatment, is used as fertilizer in the household level.
	No discharge into natural water resources in the project scenario[10] in the animal farm level.
The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C;	The NBP project activity is located in Cambodia, where is the climatic conditions are tropical with an average temperature of 27◦C with little variation in the provinces[11]
In the baseline scenario, the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m;	At household level, in the baseline scenario, the retention time was greater than one month
	At animal farm level, in the baseline scenario, the retention time was greater than one month and anaerobic lagoons had deeper lagoons than 1 m. This is observed in the baseline survey and it will be confirmed by ex-post baseline and monitoring survey.
The baseline scenario should not involve methane recovery and destruction by flaring or combustion for gainful use.	No recovery and destruction by flaring or combustion occurred in the baseline scenario for both household and animal farm level.

[9] Law on Environmental Protection and Natural Resources Management and the Sub-Decree on Water Pollution Control (1999);
https://cdc.gov.kh/wp-content/uploads/2022/04/Sub-Degree-27-on-Water-Pollution-Control_990406.pdf
[10] Law on Environmental Protection and Natural Resources Management and the Sub-Decree on Water Pollution Control (1999);
https://cdc.gov.kh/wp-content/uploads/2022/04/Sub-Degree-27-on-Water-Pollution-Control_990406.pdf
[11] https://climateknowledgeportal.worldbank.org/country/cambodia/climate-data-historical

The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project developer can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply.	For households application, some models build with direct feeding process some built with short storage (less than a week) and manual feeding process where the storage, animals and digesters are just next to each other. For animal farm application, the manure directly feeds the lagoon digester in the farms, no storage occurs.
A technical measure to ensure that the gas holding capacity of the biodigester is sufficiently large to capture the biogas during periods of non-usage. A justification to demonstrate compliance with this requirement pertaining to the biogas digester size shall be included in the Project Design Documentation (PDD)	Biodigester capacities are sufficiently large to capture biogas during periods of non-usage. Biodigester size and capacity shared[12] for both household and animal farm level.
The activity is implemented by a project developer and can include additional project participants listed in Appendix 2 of the PDD template. The individual households may be represented collectively by community organisations, etc., but do not individually act as project participants	The project only implemented by the project developer, NBP. No additional project participant occurred.
The developer must design incentive mechanism(s), which should be effective as fast as possible, for the displacing the use of inefficient baseline stoves or cooking practices by the project cooking devices for daily usage and describe the incentive mechanism(s) in the PDD/VPA-DD at the time of validation	To promote the installation of biodigester for reducing the use of inefficient baseline stove or cooking practices in household level, NBP has developed marketing activities at both national level (NBP webpage http://nbp.org.kh/; social media (https://www.facebook.com/nbp.org.kh)) and local level (local stakeholder consultation, small group meetings), the awareness raising of

[12] ER spreadsheet,"Project_Scale" tab cell E28:E34

	the benefits of using this technology benefits including financial, social and environmental benefits are well elaborated. For example, its convenience, saving from purchasing biomass fuel, avoiding health issues causing by using biomass fuel, etc. Furthermore, for domestic biodigester, NBP has provide subsidy and or monthly installment for households who install biodigester. This would encourage the users to discontinue or reduce the utilization of its baseline technology. However, regardless users stop completely or partially using its baseline technology, the estimation of emission reduction will take into account for all cases by conducting the field test for both project and baseline household. This will capture the amount of fuel used in both scenario and the differences will be used to estimate the emission reduction.
To avoid double counting or double claiming, the project developer must: a. clearly communicate its ownership rights and intention of claiming the emission reductions resulting from the project activity to the following parties by contract or clear written assertions in the transaction paperwork: all other project participants; project technology manufacturers; and retailers of the project technology; and b. inform and notify the end users that they cannot claim emission reductions from the project, and c. exclude from the project activity, any biodigester and cookstoves that are included in any other voluntary market or CDM or Article 6 based	A clause on the VER ownership is included in each sale contract form 03 between the household and the biodigester installer for the household level. The same statement is included in the form 12 (inventory report of animal farm) for covered lagoon biodigester in the animal farm level. The project boundary encompasses the geographical sites all the units commissioned from 1/07/2021 of all biodigester under the project. All biodigesters installed under NBP have a unique registration number. That number indicates when and where the digester was built, based on that numbers double counting can be avoided. For the household level, double counting with GS751 is avoided by maintaining one database

mechanisms project activity/PoA and strive not to displace the cooking devices of another CDM or voluntary project/PoA. See data and parameters not monitored (section 3.11), Avoidance of double counting or double claiming with other mitigation actions (BGTA 2), for details on this demonstration.	with both projects where all units installed after 30/06/2021 (thus starting on 01/07/2021) belong to GS11138 and all those built prior, belong to GS751. For farm biodigester (covered lagoon biodigester) a separated database is developed and managed by NBP.

Applicability criteria for CDM Methodology, AMS-I.F v5.0, which applied in animal farm level for hybrid generator application listed below in the table:

Eligibility criteria	**Assessment**
This methodology is applicable for project activities that: a) Install a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity (Greenfield plant); b) Involve a capacity addition, c) Involve a retrofit of (an) existing plant(s); or d) Involve a replacement of (an) existing plant(s).	a) This project activity is installment of new biogas based renewable energy plant installment in the animal farms. Electricity replacement is not applicable for Household application.
Project displaces grid electricity consumption (e.g. grid import) and/or captive fossil fuel electricity generation at the user end (excess electricity may be supplied to a grid)	In the animal farm level ,project displaces grid electricity consumption by mostly hybrid generators produced half of the electricity need by biogas and diesel.
Illustration of respective situations under which each of the methodology (AMS-I.D., AMS-I.F. and AMS-I.A.) applies is included in below Table 4.	The project activity applies methodology AMS-I.F, since it is contributing the type 2 in the table. The project supplies electricity to the project site area by hybrid generator or pure biogas generators where the electricity

	consumed by the national grid in the baseline scenario.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable, as the project does not involve capacity addition.
In the case of retrofit or replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement unit shall not exceed the limit of 15 MW	Not applicable, as the project does not include any retrofit or replacement activities.
If the unit added has both renewable and nonrenewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the unit added co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW	In the animal farm level ,the project introduces hybrid generator running with a mix of biogas and diesel or purely biogas running generator. The size of biodigester range from 100 kW to 1,000kW which is lower than the cap value of 15 MW.
Combined heat and power (co-generation) systems are not eligible under this category.	Project does not include combined heat & power (i.e. co-generation) activities
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power	Not applicable as the project is not a hydro power plant

density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m2.	
If electricity and/or steam/heat produced by the project activity is delivered to a third party, i.e. another facility or facilities within the project boundary, a contract between the supplier and consumer(s) of the energy will have to be entered that ensures that there is no double counting of emission reductions.	Not applicable. In this project, produced electricity is just used for internal consumption for the animal farms, which is for evaporative cooling systems and lightning.
In the case the project activities utilizes biomass, the "TOOL16: Project and leakage emissions from biomass" shall be applied to determine the relevant project emissions from the cultivation of biomass and the utilization of biomass or biomass residues.	Not applicable, there is no activity utilizes biomass in this project.

Table 4 Applicability of AMS-I.D, AMS-I.F and AMS-I.A based on project types

	Project type	AMS-I.A	AMS-I.D	AMS-I.F
1	Project supplies electricity to a national/regional grid		√	
2	Project displaces grid electricity consumption (e.g. grid import) and/or captive fossil fuel electricity generation at the user end (excess electricity may be supplied to a grid)			√
3	Project supplies electricity to an identified consumer facility via national/regional grid (through a contractual arrangement such as wheeling)		√	

4	Project supplies electricity to a mini grid system where in the baseline all generators use exclusively fuel oil and/or diesel fuel			√
5	Project supplies electricity to household users (included in the project boundary) located in off grid areas	√		

Applicability criteria for CDM Tool 3: Tool to calculate project or leakage CO2 emissions from fossil fuel combustion, V03.0:

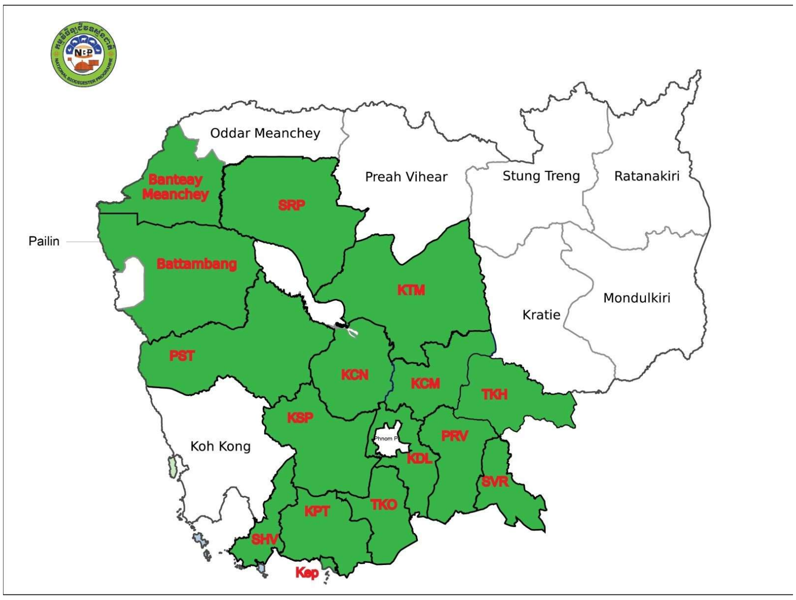
Methodologies using this tool should specify to which combustion process j this tool is being applied.

Eligibility criteria	**Assessment**
Methodologies using this tool should specify to which combustion process j this tool is being applied.	The CO_2 emissions from on-site consumption of fossil fuels by the project activity with stationary diesel combustion in the hybrid generators

B.3. Project boundary

>>

The project boundary encompasses the geographical sites of all the units commissioned from 01/07/2021 of all biodigester under the project in both household biodigester and farm biodigester. The project technologies have been installed progressively in the project target province as shown in below map and its GPS coordination.



#	Province	Latitude (xx° xx' xx" N)	Longitude (xx° xx' xx" E)
1	Kampong Cham	11° 59' 00" N	105° 27' 00" E
2	Kandal	11° 78' 30" N	104° 81' 70" E
3	Svay Reang	11° 05' 00" N	105° 48' 00" E
4	Takeo	10° 59' 00" N	104° 47' 00" E
5	Kampong Speu	11° 27' 00" N	104° 30' 00" E
6	Kampong Chhnang	12° 00' 00" N	104° 30' 00" E
7	Kampot	10° 36' 00" N	104° 10' 00" E
8	Kampong Som	10°38’00” N	103°30’00” E
9	Kep	10°29’00” N	104°18’00” E
10	Prey Veng	11°29' 00" N	105°19' 00" E
11	Siem Reap	13°21’44” N	103°51’35”E
12	Pursat	12°32’00” N	103°55’00” E
13	Battambang	13°06’00” N	103°12’00” E
14	Kampong Thom	12°42’00” N	104°53’00” E
15	Tbong khmum	11°54’34”N	105°38’49”E
16	Phnom Penh	11°33’43” N	104°53’18” E
17	Banteay Meanchey	13°45’00” N	103°00’00” E

Household biodigester: NBP provide at least three types of household biodigester including Farmer’s Friend model, NBP S1’s model and composite model with different size from 2m^3 to 15m^3 as illustrated in Table 3.In this level biodigesters feed with animal manure, optionally toilet, which connects to biogas stove, optionally lamp, and produce organic fertilizer for soil quality improvement as shown in Figure 2 below.

Farm biodigester: NBP is cooperating with private sector to promote and install “covered lagoon biodigester” at animal farm as illustrated in Figure 3 with lagoon design in Figure 4. In this level, covered lagoon biodigesters feed with direct flow from the animal barns, optionally with pump, produced biogas is fed into hybrid or pure biogas generator to produce electricity which supplies the evaporative coolers by internal electricity connection. There is a backup connection to the national electricity grid, and hybrid generators can also powered only by diesel fuel. In the event of generator stoppage, covered lagoons have enough capacity to hold the gas however flaring systems are optionally available.

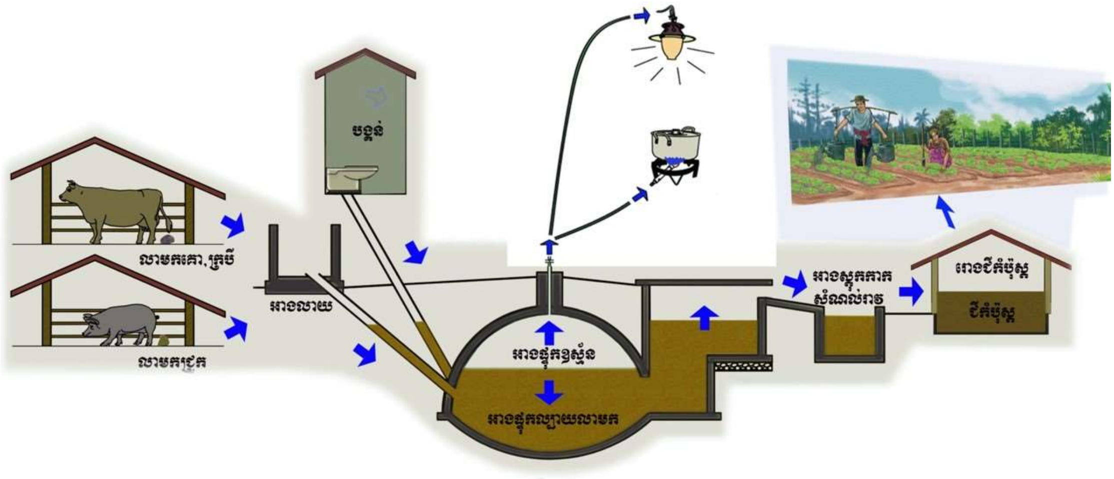


Figure 2Household Biodigesters application boundary

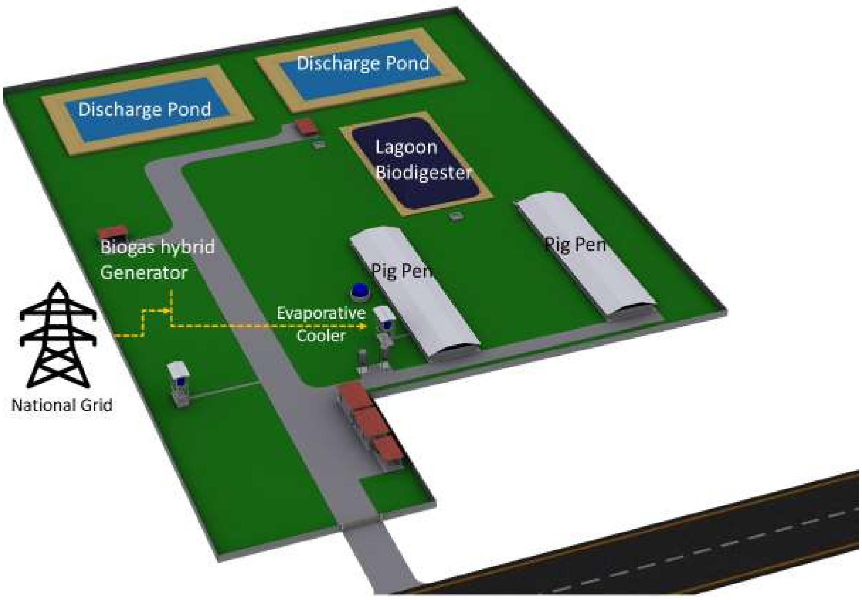


Figure 3 Animal farms biodigester and its application

Figure 4 Covered Lagoon Design

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Thermal energy demand for human food preparation and water boiling in household level	CO_2	Yes	Major source of GHG emission
		CH_4	Yes	Major source of GHG emission
		N_2O	Yes	Major source of GHG emission
	Animal waste handling and storage	CO_2	No	Excluded as CO_2 emissions from animal waste are CO_2 neutral
		CH_4	Yes	Major source of emissions
		N_2O	No	Excluded for simplification; conservative
	Generation mix of electricity grid in Cambodia , in animal farm level	CO_2	Yes	Major source of GHG emission
		CH_4	No	Minor source of GHG emission
		N_2O	No	Minor source of GHG emission
Project scenario	Biodigester system	CO_2	No	Excluded as CO_2 emissions from bio-slurry are CO_2 neutral
		CH_4	Yes	Emissions from physical leakage
		N_2O	No	Excluded as a biodigester does not produce N_2O gasses
	Thermal energy demand for human	CO_2	Yes	Major source of GHG emission
		CH_4	Yes	Major source of GHG emission

	food preparation and water boiling in household level	N_2O	Yes	Major source of GHG emission
	Diesel Fuel electricity generation with Hybrid Generator in animal farm level	CO_2	Yes	Major source of GHG emission
		CH_4	No	Minor emission source
		N_2O	No	Minor emission source

B.4. Establishment and description of baseline scenario

>>

For household biodigester, there are two baseline scenarios:

(1) The typical baseline fuel consumption patterns in a population that is targeted for adoption of the project technology, this includes households that rely mainly on wood, charcoal, and LPG which determined by the baseline survey. In the absence of the project activity times an emission factor for the fossil fuel and non-renewable biomass displaced used to calculated in details below section B.6.1, SDG13 where the main GHG emissions sources of for thermal activity can be seen in the table above ;

(2) In the baseline scenario, the Animal Waste Management System (AWMS) practices is unhygienic and the lack of access to basic sanitation result in pollution, foul odor, methane emissions and a relatively high prevalence of hygiene related diseases, such as diarrhea. In baseline, no (AWMS) practices applicable, and animal manure is left to decay anaerobically which determined by the baseline survey within the project boundary and methane is emitted to the atmosphere.

For farm biodigester (covered lagoon biodigester), there are two baseline scenarios as well:

(1) In animal farm level As per AMS-I.F., Baseline emissions are the product of amount electricity displaced with the electricity produced by hybrid or pure biogas generator and national grid emission factor with backup diesel generator. The grid electricity is the main source of energy supply at the animal farm for water pumping, lighting and evaporative cooling system with backup of diesel generator in the baseline

(2) At animal farm, the manure management is limited. Although the Ministry of Agriculture Forestry and fisheries (MAFF) issues Prakas No549 on 12 Dec 2018 by mandating the treatment of animal waste (covering lagoon) at the farm, the application of biogas generating from the treatment system is still lacking. It is observed that most of the farms just cover the lagoon while the generated biogas is released into the atmosphere without flaring or using as fuel for generating electricity or thermal. So in animal farms no Animal Waste Management System (AWMS) practices, and animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere in the baseline.

Baseline fuel use and animal waste management

As stated in the above section B.3, the sources of baseline emission are thermal demand and animal waste management.

The baseline fuel scenario is defined by the typical fuel consumption for household cooking among the target population prior to adopting the project technology. Other uses of biogas, such as electricity generation or displacement of electricity by, for example biogas water heaters, is only practiced by a minor part of the biogas population. Emission reductions arising from electricity generation are not accounted for however, this is conservative.

A Carbon monitoring survey (CMS) was executed in 2022 for GS751 CP3-MP3, in that project 4 baseline scenarios were identified, see below:

Table 5 Baseline Fuel Scenarios

Baseline scenario	Scenario [13]	Ratio [14]
Households with main fuel wood	$B_{b1,wood}$	$B_{b1,ratio}$
Households with main fuel charcoal	$B_{b2,charcoal}$	$B_{b2,ratio}$
Household with main fuel LPG	$B_{b3,LPG}$	$B_{b3,ratio}$
Housholds using other fuels	n/a	n/a

For animal waste management, in the household level it was observed that both cow and buffalo manure are mainly managed as disposed in the fields (pasture) and as solid storage, while pig manure is mostly managed by solid storage, slurry, anaerobic Lagoon and dry lot being at 36.17%, 31.74% 11.19% and 10.13% respectively as shown in Figure 5

For animal waste management, in the animal farm level according to animal farm baseline survey[15], 100% of animal manure directly left into lagoon for decay anaerobically.

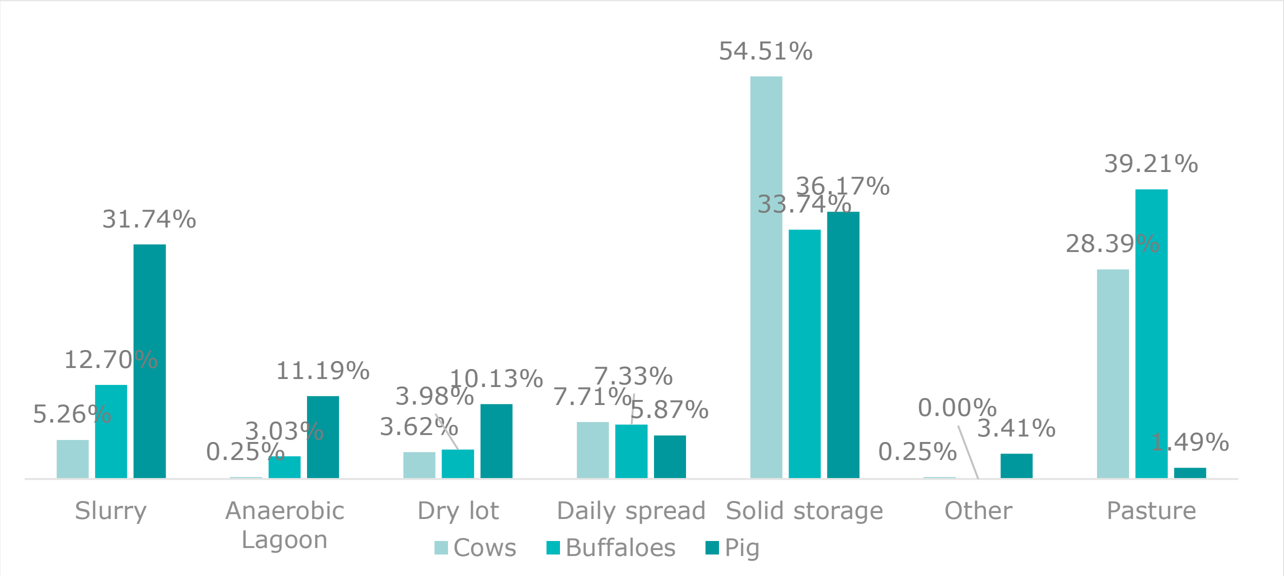


Figure 5 Baseline of animal waste management and storage for household biodigester (NBP&ATEC 2018 baseline survey data)

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[13] For wood and charcoal CP3MP1 CMS used, for LPG CP3MP2 CMS is used
[14] For each scenario latest ratio study is used from CP3MP3
[15] NBP_Animal Farm Baseline and Project Scenario Survey_2023

As stated in the applied GS methodology, for the application of biodigester, the baseline scenario does not include only the type of fuel consumption but also the animal waste management system (AWMS) in the project's target area. Thus, the emission of the baseline scenario would be coming from two main sources (1) fuel consumption for cooking/water boiling which is thermal application at household level and (2) the GHG emission from AWMS application for households and animal farm application. Additionally, per CDM methodology, the other baseline scenario is coming from grid electricity consumption where the half of need displaced by the hybrid generators from electricity application at animal farm level.

$$BE_y = BE_{th,y} + BE_{awms,y} + BE_{elec,y}$$

Where:

BEy	=	Total baseline emissions in the pre-project situation (tCO2e/year)
BEth,y	=	Baseline emissions from household's fuel consumption for thermal energy needed (tCO2e/year)
BEawms,y	=	Baseline emission from animal waste handling in household and animal farms (tCO_2e/year)
BEelec,y	=	Baseline emission from grid electricity consumption displacement in animal farms (tCO_2e/year)

B.5. Demonstration of additionality

a. Demonstration of additionality for household biodigester

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable)	**COMMUNITY SERVICES ACTIVITY requirements v1.2 Annex B – Positive list**
Describe how the proposed project meets the criteria for deemed additionality	The project installs biogas plants, As per Annex B these technologies are deemed automatic additional 1.1.2, provided the plants are less than 100 kW. As demonstrated in section ER

	Spreadsheet[16], the largest plant has a thermal output of 37.43 kW which is smaller than the threshold

b. Demonstration of additionality for animal farm biodigester:

i. Animal Waste Management Systems (AWMS) Application:

According to applied GS Methodology, AWMS Application in Animal farms falls under the COMMUNITY SERVICES ACTIVITY category where, AWMS play a pivotal role in managing animal waste and manure from animal farms in an environmentally sustainable manner. While these systems operate within the animal farm setting, their impact extends beyond individual households, positively influencing the surrounding community through better waste management practices, reduced environmental pollution, and potential enhancements in health and sanitation conditions for neighboring areas. Therefore, despite operating within farm boundaries, AWMS is recognized as a community service due to its wider-reaching benefits and contributions to communal well-being within the surrounding areas.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable)	**COMMUNITY SERVICES ACTIVITY requirements v1.2 Principle 5 – Financial Additionality & Ongoing Financial Need**
Describe how the proposed project meets the criteria for deemed additionality	According to 4.1.9 b) Community Services Activity requirements the project located in Cambodia which is LDC (Least Developed Country) by UN Office [17] deemed automatic additional.

ii. Grid Electricity Replacement by biogas application:

According to applied CDM methodology and CDM Demonstration of additionality of small-scale project activities (Tool21)[18], criteria listed below per appendix of Tool 21 applied to demonstrate additionality of Electricity application in the animal farm levels

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[16] Tab Project_Scale, cell F34
[17] https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf
[18] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf

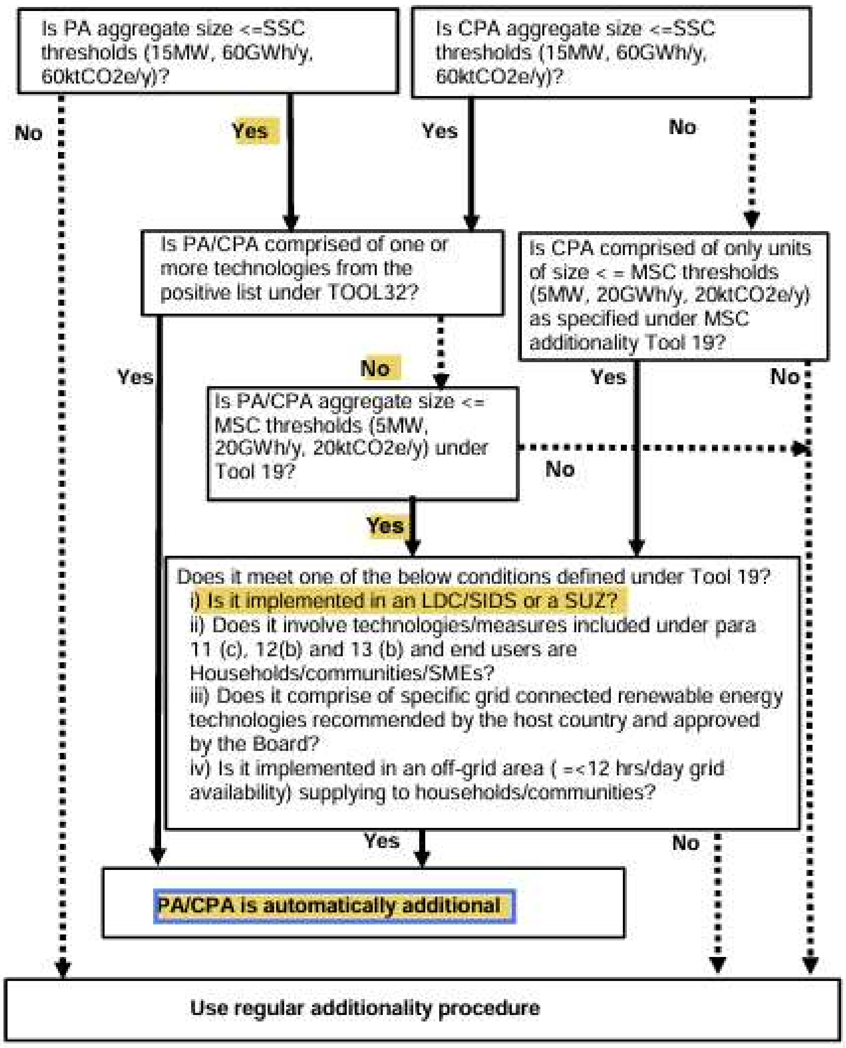

Figure 6 Criteria for automatic additionality using provisions of small-scale (SSC) or microscale (MSC) additionally tools

B.5.1 Prior Consideration

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This project is retroactive because the LSC could not be organized due to COVID-19 restrictions on gathering of groups and travel in the country.

The submission of the project for preliminary review was on 03/04/2021 which was before the project start date of 01/07/2021. Therewith prior consideration is demonstrated.

The prior consideration of revenues:

At NBP, the carbon revenue has been the main source for operating the program as reported in its annual report[19] 2019, 2020, 2021, 2022 in the below table. Furthermore, recognizing the importance of carbon revenue to sustain the operation of the program, the NBP's steering committee started discussing on hiring carbon expert to study on the establishment of the new carbon project since Nov 2019 prior to the project start date of July 2021 as shown in the minute of the steering committee meeting[20].

Year	Total expenses (USD)	Source of expenses budget from VER's (USD)
2019	1,375,930	907,888 (66%)
2020	816,620	454,949 (56%)
2021	574,148	361,647 (63%)
2022	568,364	491,584 (87%)

B.5.2 Ongoing Financial Need

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Not applicable as this information need only be included at Design Certification Renewal.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the six SDGs

SUSTAINABLE DEVELOPMENT GOALS TARGETED	MOST RELEVANT SDG TARGET	SDG IMPACT **INDICATOR (PROPOSED OR SDG INDICATOR)**
13 Climate Action (mandatory)	13.3 Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning	The total amount of emission reduction.

[19] NBP's annual report 2019, 2020, 2021, 2022
[20] Minute of the steering committee meeting

1 No Poverty	1.1 by 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than $1.25 a day	Financial savings from avoided biomass purchase
2 Zero hunger	2.4 By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, that help maintain ecosystems, that strengthen capacity for adaptation to climate change, extreme weather, drought, flooding and other disasters and that progressively improve land and soil quality	Total area on which bio-slurry is applied
7 Affordable and Clean Energy	7.1 by 2030, ensure universal access to affordable, reliable and modern energy services	Household access to clean energy with the biodigesters sold under the project activity
	7.2 By 2030, increase substantially the share of renewable energy in the global energy mix	Electricity consumption by users produced by the project Renewable energy (MWh)
8 Decent Work and Economic Growth	8.1 Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries	Number of jobs created by the project activity
15 Life on Land	15.2 by 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally	Forest area saved

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

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Goal 1 contribution

The project technology help users to save household expenditure on fuel purchased for cooking and saved time spent for collecting and/or time for buying wood and LPG.

The indicator for this SDG1 would be Financial savings from avoided biomass consumption, which is relevant to the UN's SDG indicator "1.1.1 Proportion of population below the international poverty line, by sex, age, employment status and geographical location (urban/rural).

Estimating baseline outcome:

In the baseline situation, no fuel saving. Therefore, the baseline outcome benefits are

zero.

Estimating project outcome:

In the project situation, the project outcome will be estimated based on the saved biomass and LPG fuel which will be surveyed during monitoring survey.

Project outcome of SDG1

Financial savings from avoided biomass purchase = $N_{p,unit\text{-}HH,y}$ *$U_{p,y}$ * ($P_{b,wood,y}$ – $P_{p,wood,y}$)* Biomass Price ($ per ton)

Estimating net benefit:

Net benefit of SDG1 = Project outcome of SDG1 – Baseline outcome of SDG1

For ex-ante estimation of SDG contribution, the Ex-ante value is estimated to be

equal to the latest baseline survey result:

Financial savings from avoided biomass purchase= 1,489 ($N_{p,unit\text{-}HH,y}$ monthly average biodigester number)*0.81 ($U_{p,y}$ usage rate)* 0.94 tonnes/year ($P_{b,wood,y}$- $P_{p,wood,y}$)*0.075$ [21] (Biomass Price per kg) *1000 tonnes/kg (Conversion factor)

= 85,792$

[21] HSE Monitoring Survey 2021

Goal 2 contribution

Bio-slurry granular fertilizer enhance soil productivity because of the use of bio-slurry (the bio-slurry granular fertilizer contained N=1.12, P=5.37, K=7.02 and OM =20.41%.) The use of granular production benefits to increase production cultivation, and dissemination on effectiveness usage

Estimating baseline outcome**:** In the baseline no construction of biodigesters occurred. Therefore, baseline outcome benefit is zero.

Estimating project outcome: The contribution to this SDG is the usage of bio-slurry as an organic and climate friendly input to sustainable agriculture. Total area on which bio-slurry is applied calculated with the following equation:

Equation 1: Calculation of contribution to SDG2

$$A_{bs,y} = N_{p,unit,y} \times U_{p,y} \times \%BS_y \times A_{bs,h,y}$$

Where:

$A_{bs,y}$	=	Total area on which bio-slurry is applied in year y in hectare
$N_{p,unit,y}$	=	Monthly Average Number of total Biodigesters installed in year y
$U_{p,y}$	=	Usage rate in year y
$\%BS_y$	=	Share of households that use bio-slurry for crop production in year y
$A_{bs,h,y}$	=	Average area per household on which bio-slurry is applied in year y

Estimating net benefit:

Net benefit of SDG2 = Project outcome of SDG2 – Baseline outcome of SDG2

For ex-ante estimation of SDG contribution, the Ex-ante value is calculated with equation 2 total area on which bio-slurry is applied

$$A_{bs,y} = 1489(monthly\ average\ biodigester\ number) * 0.81(usage\ rate) * 0.97(bioslurry\ usage\ rate) * 0.69(ha/yr)$$

= 812 ha/ yr

Goal 7 contribution:

The project contributes SDG7 in two different ways. For the household application SDG7(a) Household access to clean energy with the biodigesters sold under the project activity , SDG7(b) Electricity consumption by users produced by the project Renewable energy

SDG7(a): The project technology provides improved bio-digesters to the users which is contributing to the target "7.1 by 2030 ensure universal access to affordable, reliable, and modern energy services". The SDG's indicator of this target is the Number of household access to clean energy by biodigesters sold under the project activity which is relevant to the UN's SDG indicator "7.1.2 Proportion of population with primary reliance on clean fuels and technology."

SDG7(b): For the animal farms application where the grid electricity consumption replaced with hybrid generator contributes target "7.2 by 2030, increase substantially the share of renewable energy in the global energy mix". The SDG's indicator of this target is the MWh of net electricity displaced under the project activity which is relevant to the UN's SDG indicator "7.2.1 Renewable energy share in the total final energy consumption."

Estimating Baseline outcome:

In baseline situation, no distribution of project technology. Therefore, baseline outcome benefit is zero.

Estimating Project outcome:

In the project situation, the Number of households access to clean energy by number of biodigester sale in year y ($N_{p,biodigester\text{-}HH,y}$) under the project activity will be monitored by the project sales database .

Electricity consumption by users produced by the project Renewable energy calculated by survey inputs of electricity bills.

Where yearly average of grid electricity replacement is 89.7 MWh, calculated by baseline and project scenario grid electricity consumption bills. This amount is determined by the official bills. This difference is replaced by hybrid and pure biogas generators.

Quantity of net electricity displaced as a result of the implementation of the project activity in year y (MWh/yr) – $EG_{BL,y}$ is determined depending on replaced amount of grid electricity consumption with usage of hybrid generators can be used by taking account that the average of 70%[22] production by the hybrid and pure generator supported by the biogas which gives 62.8 MWh/year replacement as average.

Estimating the net benefit

For Household application the net benefit of SDG7 calculates as:
Net benefit of SDG7 = Project outcome of SDG7 – Baseline outcome of SDG7
SDG7(a) contribution = $N_{p,biodigester\text{-}HH,y}$ = 667 (biodigester/year) = 667 Households

For the animal farms application, the net benefit of SDG7 calculates as:
Net benefit of SDG7 = Baseline outcome of SDG7 - Project outcome of SDG7
SDG7(b) contribution = 62.8 MWh/year[23] ($EG_{BL,y}$) * 50 farms/year ($N_{p,units\text{-}afarms,y}$) = 3,141 MWh

Goal 8 contribution:

National Biodigester Programme, creates job opportunity with the project activity.
NBP full-time Staffs are based in Phnom Penh and work around in targets provinces but also create job in MAFF level as counterpart staff and PBP level as staff. .

[22] NBP_Animal Farm Baseline and Project Scenario Survey_2023
[23] ER spreadsheet, tab ER_elec, cell P35

Through the project's activities, it will create jobs which contribute to the target 8.1 "Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries". Number of jobs created is used as indicator of this SDG8, which is relevant to the UN's SDG indicator.

An estimate of jobs created by the project activities will be taken from annual reports.

Estimating baseline outcome:

In the baseline situation, no new jobs will be created. Therefore, baseline outcome benefit is zero.

Estimating project outcome:

In the project situation, number jobs created ill be recorded by the project implementer:

Project outcome of SDG8 = Number of Jobs Created

Estimating net benefit

Net benefit of SDG8= Project outcome of SDG8 – Baseline outcome of SDG8

For ex-ante estimation, it is expected that more than 80 jobs will be created by the dissemination of the biodigesters

SDG 13 contribution

According to the selected methodologies, the project will help to save fossil fuel non-renewable biomass and improve animal waste management practices also reduce grid electricity consumption which therefore reduce the GHG emission. The amount of ER will be calculated according to the selected methodologies. The following section will describe a step by step in estimating ER. The Gold Standard Baseline and Monitoring Methodology Biodigester v1. used to calculate thermal and AWMS application and CDM Methodology AMS-I.F.: Renewable electricity generation for captive use and mini grid-V5.0 used to calculate displaced grid electricity consumption.

1. Avoidance of methane emissions from AWMS
2. Displacement of non-renewable biomass and fossil fuels
3. Electricity Application grid displacement

The overall emission reduction from the project can be calculated as following:

$$\mathbf{ER_y = N_{p,unit,y} * (\underbrace{ER_{th,y}}_{i} + \underbrace{ER_{AWMS,y}}_{ii} + \underbrace{ER_{elec,y}}_{iii})}$$

Equation 2

Where

ER_y	Total emission reduction per year (tCO2e/year)	
$N_{p,unit,y}$	Monthly Average Number of total biodigesters installed in year y	
$ER_{th,y}$	Emission reduction generating from displacement of thermal energy (cooking) (tCO2e/year)	i
$ER_{AWMS,y}$	Emission reduction generating from improving animal waste management system (AWMS) (tCO_2e/year)	ii
$ER_{elec,y}$	Emission reduction from displacement of grid electricity consumption (tCO2e/year)	iii

i. Emission reduction generating from displacement of thermal energy (tCO_2e/year)

Emission reductions are credited by comparing fuel consumption in a project scenario to the three baseline scenarios. As the baseline fuel and the project fuel and the corresponding emission factors are different, the overall GHG reductions achieved in year y for thermal energy calculated as follows:

$$\mathbf{ER_{th} = (BE_{th,hh,y} - PE_{th,hh,y}) * 0.95}$$

Equation 3

Where

$BE_{th,hh,y}$	Baseline Emission generating from the use of thermal energy (tCO_2e/year)	(i-a)
$PE_{th,hh,y}$	Project Emission generated by the use of thermal energy (tCO_2e/year)	(i-b)
0.95	adjustment factor approximate leakage emissions for thermal application.	(i-c)

i-a. Estimating baseline emission generating from the thermal use

According to Gold Standard methodology Thermal application method 1: Based on avoided quantity of fuel consumption used to calculate baseline emissions as follows;

Equation 3

$$BE_{th,hh,y} = \Sigma_{b,p}(N_{b,p,y} * U_{p,y} * f_{NRB,i,y} * (SE_{b,y,CO2} + SE_{b,y,nonCO2}))$$

Where

$BE_{th,hh,y}$	Baseline Emission generating from the use of thermal energy (tCO_2e/year)
$\Sigma_{b,p}$	Sum over all relevant baseline b/project p pairs
$N_{b,p,y}$	Number of project technology-days included in the project database for each project scenario in year y.

$U_{p,y}$	Usage rate for technologies in project scenario p in year y (fraction)
$SE_{b,y,CO2}$	Specific CO2 emissions for a baseline b technology in year y (tCO2/technology*day)
$SE_{b,y,nonCO2}$	Specific non-CO2 emissions for a baseline b technology in year y (tCO2e/technology*day)
$f_{NRB,i,y}$	Fraction of biomass used during year y for the considered scenario b that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario)

- **Specific Emission per Fuel type (SE_{CO2} & $SE_{non\text{-}CO2}$) calculation**

$$SE_{b,y,CO2} = \Sigma\, P_{b,i,y} * NCV_{b,i,fuel} * EF_{b,i,CO2}$$
$$SE_{b,y,non\text{-}CO2} = \Sigma\, P_{b,i,y} * NCV_{b,i,fuel} * EF_{b,i,non\text{-}CO2}$$

Equation 4

Where:

i	Index for the type of baseline/fossil fuel consumed
$P_{b,i,y}$	Average yearly consumption of baseline fuel i per household before the start of the project activity or at the renewal of each crediting period, whichever is later (tonnes/household/year)
$NCV_{b,i,fuel}$	Net calorific value of the fuel(s) i that is substituted in baseline b (TJ/tonne)
$EF_{b,i,CO2}$	CO2 emission factor arising from use of fuels i in baseline scenario (tCO2/TJ)
$EF_{b,i,non\text{-}CO2}$	Non-CO2 emission factor arising from use of fuels in baseline scenario (tCO2e/TJ)

Baseline Emission per Fuel type (Wood , Charcoal , LPG) calculation

$$BE_{th,hh,wood,y} = N_{b,p,y,} * U_{p,y} * ((ʆ_{NRB,i,y} * P_{b,wood,y} * NCV_{b,wood} *(EF_{b,wood,CO2}+ EF_{b,wood,non\text{-}CO2}))$$

$$BE_{thhhcharcoal,y} = N_{b,p,y,} * U_{p,y}* ((ʆ_{NRB,i,y} * P_{b,charcoal,y} * NCV_{b,wood} *(EF_{b,charcoal,CO2}+ EF_{b,charcoal,non\text{-}CO2}))$$

$$BE_{thhhLPG,y} = N_{b,p,y,} * U_{p,y} * ((ʆ_{NRB,i,y} * P_{b,LPG,y} * NCV_{b,LPG} *(EF_{b,LPG,CO2}+ EF_{b,LPG,non\text{-}CO2}))$$

Equation 5

Baseline Fuel Test (BFT)

BFT is conducted before the 1st verification of the project. Wood and charcoal fuel consumption will be measured with a representative sample of potential technology users to measure the amount of wood and or charcoal use in the baseline scenario. The emission from fuel use will be calculated as per option 2 of the methodology. LPG consumption will not be monitoring using the BFT but instead determined using survey methods for safety purposes, see section B.7.1 for the monitoring procedure of LPG.

Fuel type	$N_{b,p,y}$ (days)	$U_{p,y}$ (%)	$f_{NRB,i,y}$ (%)	$P_{b,i,y}$ Average per HH (ton/year)	$EF_{b,i,CO2}$ (tCO2/TJ)	$EF_{b,i,nonCO2}$ (tCO2e/TJ)	$NCV_{b,i,fuel}$ (TJ/ton)	Thermal energy demand (tCO2/year)
Firewood	365	81.37%	95%	1.0859	112	9.46	0.0156	1.59
Charcoal	365	81.37%	95%	0.0147	112	5.865	0.0295	0.04
LPG	365	81.37%	100%	0.0153	63.1	0.960	0.0473	0.04
							Total	1.67

i-b. Project emission generating from the use of thermal energy

According to Gold Standard methodology Thermal application method 1: Based on avoided quantity of fuel consumption used to calculate project emissions as follows:

$$PE_{th,hh,y} = \Sigma_{b,p} (N_{b,p,y} * U_{p,y} * f_{NRB,y} * (SE_{p,y,CO2} + SE_{p,y,non\text{-}CO2}))$$

Equation 6

Where

$PE_{th,hh,y}$	Project Emission generated by the use of thermal energy (tCO2e/year)
$\Sigma_{b,p}$	Sum over all relevant baseline b/project p pairs
$N_{b,p,y}$	Number of project technology-days included in the project database for each project scenario in year y
$U_{p,y}$	Usage rate for technologies in project scenario p in year y (fraction)
$SE_{p,y,CO2}$	Specific CO2 emissions for a project p technology in year y (tCO2/technology*day)
$SE_{p,y,non\text{-}CO2}$	Specific Non-CO2 emission for a project p technology in year y (tCO2e/technology*day)
$f_{NRB,i,y}$	Fraction of biomass used during year y for the project scenario p that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario)

- **Specific Emission per Fuel type (SE_{CO2} & $SE_{non\text{-}CO2}$) calculation**

$$SE_{p,y,CO2} = \Sigma\, P_{p,i,y} * NCV_{p,i,fuel} * EF_{p,i,\ CO2}$$
$$SE_{p,y,non\text{-}CO2} = \Sigma\, P_{p,I,y} * NCV_{p,i,fuel} * EF\ p_{,i,\ non\text{-}CO2}$$

Equation 7

Where:

i	Index for the type of project/fossil fuel consumed
$P_{p,i,y}$	Average daily consumption of project fuel i per household (tonnes/household/day)
$NCV_{b,i,fuel}$	Net calorific value of the fuel(s) i that is used in project p (TJ/tonne)
$EF_{p,i,CO2}$	CO2 emission factor arising from use of fuels i in project scenario (tCO2/TJ)
$EF_{p,i,non\text{-}CO2}$	Non-CO2 emission factor arising from use of fuels in project scenario (tCO2e/TJ)

Project Emission per Fuel type (Wood, Charcoal, LPG) calculation

$$PE_{th,hh,wood,y} = N_{b,p,y,} * U_{p,y} * ((f_{NRB,i,y} * P_{p,wood,y} * NCV_{p,wood} * (EFp_{,wood,CO2} + EFp_{,wood,non\text{-}CO2}))$$
$$PE_{th,hh,charcoal,y} = N_{b,p,y} * U_{p,y} * ((f_{NRB,i,y} * P_{p,charcoal,y} * NCV_{p,wood} * (EF_{p,charcoal,CO2} + EF_{p,charcoal,non\text{-}CO2}))$$
$$PE_{th,hh,LPG,y} = N_{b,p,y} * U_{p,y} * ((P_{p,LPG,y} * NCV_{p,LPG} * (EF_{p,LPG,CO2} + EF_{p,LPG,non\text{-}CO2}))$$

Equation 8

Project fuel test (PFT)

Households may continue to use NRB and LPG next to using biogas. Wood and charcoal fuel consumption will be measured with a representative sample of end users in order to measure real, observed technology performance in the field; the so-called Project Fuel Test. The emission reductions from fuel displacement will be calculated as per option 2 of the methodology for each verification. LPG consumption will not be monitoring using the PFT but instead determined using survey methods for safety purposes, see section B.7.1 for the monitoring procedure of LPG.

Fuel type	$N_{b,p,y}$ (days)	$U_{p,y}$ (%)	$f_{NRB,i,y}$ (%)	$P_{p,y}$ Average per HH (ton/year)	$EF_{b,CO2}$ (tCO2/TJ)	$EF_{p,nonCO2}$ (tCO2e/TJ)	NCV (TJ/ton)	Thermal energy demand (tCO2/year)
Firewood	365	81.37%	95%	0.142	112	9.46	0.0156	0.21
Charcoal	365	81.37%	95%	0.011	112	5.865	0.0295	0.03
LPG	365	81.37%	100%	0.019	63.1	0.960	0.0473	0.05
							Sum	0.28

i-c.Leakage emission generating from the use of thermal energy

Leakage emissions, LEp,y determined according to Option 1 per the methodology ; default adjustment factor of 0.95 to the emission reductions to approximate leakage emissions for thermal application.

Emission Reduction

The emission reductions for the thermal energy calculated as follows:

$$ER_{th} = \left(BE_{th,hh,y} - PE_{th,hh,y}\right) * 0.95$$ **Equation 9**

$= (1.67 – 0.28) * 0.95$

$= 1.31$ tCO2/year per House Hold

ii. Calculating emission reduction generating from animal waste management system (ER_{awms})

The emissions from the animal waste management system of the baseline are determined using the IPCC 2006 Tier 2 approach[24]. The approach requires information about the characteristics of the manure and its the management systems in the baseline. Depending on that survey about baseline animal manure management practices conducted. This approach is applicable for households and animal farms with distinctive animal waste management system, where the majority of the waste is collected and where the animals are kept near the houses or directly in the farms close to the covered lagoons. The following formulas are used to estimate animal waste management emissions.

$$ER_{AWMS} = BE_{AWMS,y} - PE_{AWMS,y} – LE_{AD,y}$$

Equation 10

Where

$BE_{AWMS,y}$	Baseline Emission generating from AWMS (tCO2e/year)	(ii-a)
$PE_{AWMS,y}$	Project Emission generated from AWMS (tCO2e/year)	(ii-b)

[24] Where annual emission reduction for methane recovery component is higher than five tonnes of CO2eq per biodigester the AWMS method 2 shall be applied.

$LE_{AD,y}$	Leakage emissions associated with the anaerobic digester in year y (t CO2e)

ii-a. Calculating baseline emission generating from AWMS (BEawms)

According to Gold Standard methodology AWMS method 2: IPCC TIER 2 approach (IPCC 2019) used to calculate baseline emissions as follows:

$$BE_{AWMS,h,y} = (N_{b,p,y}/365) * GWP_{CH4}*D_{CH4}*UF_b*U_{p,y}*\sum_k \sum_{j,LT} (MCF_{j,k}*B_{0,LT}*N_{LT,y}*VS_{LT,y}* MS\%_{Bl,j})$$

Equation 11

Where:

Nb,p,y	Number of project technology-days included in the project database for each project scenario in year y
GWP_{CH4}	Global Warming Potential (GWP) of CH4 applicable to the crediting period (IPCC default value 28 tCO2e/tCH4)
D_{CH4}	CH4 density (0.00067 t/m3 at room temperature (20 ºC) and 1 atm pressure)
UF_b	Model correction factor to account for model uncertainties (0.94 UNFCCC default value)
$U_{p,y}$	Usage rate for technologies in project scenario p in year y (fraction).
$MCF_{j,k}$	Annual methane conversion factor (MCF) for the baseline animal manure management system j by climate region k. (%)
$B_{0, LT}$	Maximum methane producing potential of the volatile solid generated for animal type LT (m3 CH4/kg-dm)
$N_{LT,y}$	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$MS\%_{Bl,j}$	Fraction of manure handled in baseline animal manure management system j

Country specific default data is not available for $VS_{LT,y}$, $B_{0,LT}$, $MCF_{j,k}$ and $MS\%_{Bl,j}$. 2019 Refinement to IPCC 2006 default values will be used for $B_{0,LT}$ and $MCF_{j,k}$. $MS\%_{Bl,j}$ will be obtained using survey methods (see section B.7.1).

➢ **Calculating N_{LT} for tier 2 approach**

$$N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365}\right)$$

Equation 12

Where,

$N_{da,y}$	Number of days animal is alive in the farm in the year y (numbers)
$N_{p,y}$	Number of animals produced annually of type LT for the year y (numbers)

Animal	$N_{da,y}$	$N_{p,y}$ HH	$N_{da,a\text{-}farm,y}$	$N_{p,y}$ animal farms
Cow (other cattle)	365.00	5.35	0.00	0.00
Pig	365.00	3.20	300.00	3437.50
Buffalo	365.00	0.83	0.00	0.00

$N_{LT,y}$ HH	$N_{LT,y}$ animal farms
5.35	0.00
3.20	2825.34
0.83	0.00

- **Calculating** $VS_{(LT,y)}$ **for tier 2 approach**

According to methodology default IPCC value of VS may be adjusted for a site-specific average animal weight. Due to the difficulty in measuring the weight of cattle and buffalo in the fields, only the pig's weight was estimated according to average of entry weight at the farm and weight when the pig leaves the farm. Farmers are often able to estimate the weight of fattening pigs reliably as the weight of birth is known and the value of selling the pigs is based on their weight. Based on NBP Monitoring report MPV, the average pig's weight was 54.13kg. For the other types of animals , TAM default values used according to IPCC values.

$$VS_{LT,y} = (VS_{rate,LT} \times \frac{TAM_{LT}}{1000}) \times nd_y$$

Equation 13

Where:

$VS_{rate,LT}$	Default VS excretion rate
TAM_{LT}	Typical animal mass for livestock LT (kg/animal)
nd_y	Number of days treatment plant was operational in year y

Animal (LT)	$VS_{rate,LT}$ (kg-Vs/ 1000 kg animal mass /day)	TAM_{LT} (kg/animal)	nd_y (#days)	$VS_{LT,y}$ (kg-dm/animal/yr)
Cow (other cattle)	9.80	299.00	365.00	1,069.52
Pig	6.80	54.13	365.00	134.35

Buffalo	13.50	336.00	365.00	1,655.64

$$\underbrace{N_{LT,y} \times B_{0,LT} \times VS_{LT,y} \times MCF_{j,k} \times MS\%_{Bl,j}}_{\boxed{(i)}}$$

Calculating (i)

Animal (LT)	**$VS_{LT,y}$** (kg-dm/animal/yr)	**$N_{LT,y}$ HH (# Animal)**	**$N_{LT,y}$ animal farms (# Animal)**	**$B_{0,LT}$ (m3CH4/kg-dm)**	**$MCF_{j,k}$ *$MS\%_{(Bl,j}$ (%)**	**(i) HH (m3CH4/animal /year)**	**(i)animal farms (m3CH4/animal/ year)**
Cow (other cattle)	1069.52	5.35	0.00	0.13	7.55%	56.10	0.00
Pig	134.35	3.20	2825.34	0.29	34.64%	43.15	38132.35
Buffalo	1655.64	0.83	0.00	0.10	14.48%	19.90	0.00

- **Baseline Emission per animal type calculation for households and animal farms**

Animal (LT)	$N_{b,p,y}$ (days)	GWP_{CH4} (t_{CO2}/t_{CH4})	$U_{p,y}$ (%)	D_{CH4} (t_{CH4}/m^3)	UF_b (fraction)	**(i) HH (m3CH4/animal/year)**	**(i)animal farms (m3CH4/animal/year)**	**$BE_{AWMS,HH,y}$ (tCO_2e/HH/year)**	**$BE_{AWMS, animal farm,y}$ (tCO_2e/animal farm/year)**
Cow	365	28	81.37%	0.00067	0.94	56.10	0.00	0.805	0.000
Pig	365	28	81.37%	0.00067	0.94	43.15	38132.35	0.619	547.160
Buffalo	365	28	81.37%	0.00067	0.94	19.90	0.00	0.286	0.000
							Total	**1.710**	**547.160**

ii-b. Project emission generated from AWMS (PE_{AWMS})

As stated above Project emissions from AWMS application divide into two according to project application. To calculate project emission from AWMS from households and animal farms application according to GS methodology, project emission involve emissions from the biodigester is composed of four main ways: Physical leakage via biodigester which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use. Emissions from the use of fossil fuels or electricity for the operation of all the installed facilities. Emissions from incremental transportation and Emissions from the storage of manure before being fed into the anaerobic digester. According to Gold Standard methodology AWMS method 2: IPCC TIER 2 approach (IPCC 2019) used to calculate project emissions as follows:

$$PE_{AWMS,y} = PE_{PL,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Where:

$PE_{AWMS,y}$	=	Project emission associated AWMs per h per year (tCO2e/year)
$PE_{PL,y}$	=	Emissions due to physical leakage of biogas in year y (t CO2e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO2e)
$PE_{transp,y}$	=	Emissions from incremental transportation of input manure for the digesters in the year y (t CO2e)
$PE_{storage,y}$	=	Emissions from the storage of manure (t CO2e)

For this project PE_{transp} and $PE_{storage}$ excluded since every digester located next to the animals and feed directly into digester without any storage or waiting times for the both level of application.

For household application no electricity or fossil fuel is used associated with the anaerobic digesters. For the animal farms application, PE_{power} calculated separately below on the following part iii by CDM methodology, since the animal farm application use hybrid generator with grid electricity for cooling purposes in the farms.

So finally,

$$PE_{AWMS,y} = PE_{PL,y}$$

Equation 14

According to GS methodology physical leakage of biogas in the manure management systems which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use ($PE_{PL,y}$) are estimated as 10% of the maximum methane producing potential of the manure fed into the management systems implemented by the project activity as per the following equation:

$$PE_{PL,y} = 0.10 \times N_{b,p,y} \div 365 \times U_{p,y} \times GWP_{CH4} \times D_{CH4} \times \sum_{k} \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$$

Equation 15

$PE_{PL,y}$	Emissions due to physical leakage of biogas in year y (t CO_2e)
GWP_{CH4}	Global Warming Potential (GWP) of CH_4 applicable to the crediting period
D_{CH4}	CH_4 density (0.00067 t/m^3 at room temperature (20 ºC) and 1 atm pressure)
LT	Index for all types of livestock
i,	Index for animal manure management system
k	Climate region k
$B_{o,LT}$	Maximum methane producing potential of the volatile solid generated for animal type LT
$MS\%_{i,y}$	Fraction of manure handled in project animal manure management system i

➢ **Project Emission from AWMS application in households**

Animal	IPCC default	Nb,p,y (days)	$U_{p,y}$ (%)	GWP_{CH4} (tCO2/tCH4)	D_{CH4} (tCH4/m3)	$B_{0(T)}$ (m3CH4/kg-dm)	$N_{LT,y}$ HouseHold (#animals)	$VS_{LT,y}$ (kg-dm/animal/yr)	$MS\%_{Bl,j}$ HouseHold	$PE_{PL,y}$ HouseHold
Cow	%10	365	81.37%	28	0.00067	0.13	5.35	1,069.52	49.59%	0.562
Pig	%10	365	81.37%	28	0.00067	0.29	3.20	134.35	27.64%	0.052
Buffalo	%10	365	81.37%	28	0.00067	0.10	0.83	1,655.64	46.47%	0.097
									Total	**0.713**

➢ **Project Emission from AWMS application in animal farms**

Animal	IPCC default	Nb,p,y (days)	$U_{p,y}$ (%)	GWP_{CH4} (tCO2/tCH4)	D_{CH4} (tCH4/m3)	$B_{0(T)}$ (m3CH4/kg-dm)	$N_{LT,y}$ animal farms (#animals)	$VS_{LT,y}$ (kg-dm/animal/yr)	$MS\%_{Bl,j}$	$PE_{PL,y}$ animal farm
Pig	%10	365	81.37%	28	0.00067	0.29	2825.34	134.35	100.00%	168.036
									Total	**168.036**

Leakage emissions

The project proponent should investigate the following potential sources of leakage emissions ($LE_{AD,y}$):

According to tool AWMS method 2, if digestate is being stored under anaerobic conditions, it can cause CH_4 emissions due to further anaerobic digestion of the residual biodegradable organic matter, since under the project activity no anaerobic storage occurred it is negligible. The physical leakage from the biodigesters are calculated above under the project emissions. So $LE_{AD,y} = 0$

Emission Reduction

The emission reductions for the AWMS calculated as follows:

$ER_{AWMS,HH} = BE_{AWMS,HH,y} - PE_{AWMS,HH,y}$

=1.71-0.71 = 1 tCO2/yr for each household biodigester

$ER_{AWMS,afarm} = BE_{AWMS,afarm,y} - PE_{AWMS,afarm,y}$

=547.16-168.03=379.13/yr for each animal farm biodigester

iii. Calculating emission reduction generating from grid electricity displacement

$$ER_{elec} = BE_{elec,y} - PE_{elec,y}$$

Equation 16

Where

$BE_{elec,y}$	Baseline Emission from grid electricity usage (tCO2e/year)	(iii-a)
$PE_{elec,y}$	Project Emission from fossil fuel usage (tCO2e/year)	(iii-b)

iii-a . Baseline Emission calculation for grid electricity displacement

The baseline scenario is the usage of electricity from grid connection and diesel back-up to power the cooling and lighting systems in the animal farms where farmers have to cover high expenses for fuels they displaced electricity usage from grid during the peak hours (day time).

The project will contribute towards emission reduction by reducing the greenhouse gas emissions with a fuel switch of grid connection to hybrid generator systems where the generators are supplied by biogas and diesel fuel still with grid backup, which will be calculated based on the CDM methodology AMS-I.F.

The energy generated by the project is used for pumping water to clean pigs and stables, lighting (needed both day and night), and for running the evaporative cooling system.

As per paragraph 20 of AMS-I.F. version 05.0, the baseline emissions for other systems are the product of amount electricity displaced with the electricity produced by the renewable generating unit and an emission factor.

$$BE_{elec,y} = EG_{BL,y} * EF_{CO2,y}$$

Equation 17

Where,

EG_{BLy}	Quantity of net electricity displaced as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{CO2,y}$	Emission factor (tCO2/MWh)

According to applied methodology emission factor of electricity $EF_{CO2,y}$ calculated per AMS-I.D which is $EF_{grid,CM,y}$ per Tool 07. For the value of grid electricity, data published by Institute for Global Environmental Strategies (IGES) and National Council for Sustainable Development, Cambodia (2016) is used, which is calculated by 6 different steps to determine $EF_{grid,CM,y}$ (0.389 tCO2/MWh). So the baseline emission is calculated as follows;

=89.73 MWh/ year /generator * 0.3089 tCO2/MWh

=27.71 tCO_2/ year/generator

iii-b. Project Emission calculation for grid electricity displacement

As per the approved consolidated Methodology AMS.I.F. Version 5.0 "For most renewable energy power generation project activities, PE_y = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_{elec,y} = PE_{FF,y}$$

Equation 18

$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year y (tCO_2e/yr)

As the project activity is the installation of biogas power plant into the animal farms where it replaces half of the grid connected electricity consumption with diesel & biogas hybrid generator systems, will cause project emission due to fossil fuel consumption where it calculated according to CDM Tool 3 "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion" v03.0. CO_2 emissions from fossil fuel combustion in process calculated based on the quantity of fuels combusted and the CO_2 emission coefficient of those fuels, per tool as follows:

$$PE_{\mathrm{FF},y} = FC_{diesel,generator,y} * COEF_{diesel,y}$$

Equation 19

$FC_{diesel,generator,y}$	Quantity of fuel type (diesel) combusted in process (generator) during the year y (mass or volume unit/yr)
$COEF_{diesel,y}$	CO_2 emission coefficient of fuel type (diesel) in year y (tCO_2/mass or volume unit)

Where the COEF calculated as option B according to tool as:

$$COEF_{diesel,y} = NCV_{diesel,y} * EF_{CO2,diesel,y}$$

Equation 20

$NCV_{diesel,y}$	Weighted average net calorific value of the fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,diesel,y}$	Weighted average CO2 emission factor of fuel type (diesel) in year y (tCO2/GJ)

COEF = 0.0439 (GJ/kg) * 0.0726 (tCO2/GJ) = 0.00318714 tCO_2/kg

The project emission with replaced diesel fuel amount where diesel back-up use for both baseline and project scenario replace almost one third (%30) of the grid connection from the baseline situation calculated as follows per generator as follows:

$PE_{FF,y} = FC_{diesel,generator,y} * COEF_{diesel,y} = 9720(lt/y)*0.835\ (kg/lt)$[25]$*0.00318\ tCO_2/kg = 26.14\ tCO_2/y/generator$

Emissions related fossil fuel consumption $PE_y = 26.14\ tCO_2/generator$

Emission reductions are calculated as:

$ER_y=BE_y-PE_y$

$=27.71 – 26.14$
$=1.57\ tCO_2/year$ per generator

With average number of 50 animal farms and generator per year, total ER by electricity displacement application will be 78.5 $tCO_2/year$

- The overall emission reduction from the project can be calculated from equation (1) with the total usage percentage and amount of biodigester installed, which are changing year by year. Total emission reduction per application as, household and animal farms level summarized below in the table. Detailed yearly calculation by each application type can be found in the ER excel sheet with individually SDG contribution calculations:

	BASELINE EMISSION (BE)			PROJECT EMISSION (PE)			Leakage Emission (LE)	Emission Reduction (ER)			Total per biodigester (tCO2/year)
	Thermal	AWMS	Electric	Thermal	AWMS	Electric	Thermal	Thermal	AWMS	Electric	
HouseHold	1.668	1.710	-	0.284	0.713	-	95%	0.997	1.315	-	**2.312**
Animal farms	-	547.160	58	-	168.036	56.39	-	-	379.124	1.574	**380.698**

SDG 15 contribution

The project technology help users to reduce amount of fuel which contribute to the area of forest save. The indicator for this SDG15 would be the Forest area saving which is relevant to the UN's SDG indicator "15.2 By 2020, promote the implementation of

[25] Density of Diesel fuel, used for unit conversion (Source CDM- AMS.I-B,V13.0)

sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally.”

Estimating baseline outcome:

In baseline situation, no area of forest saved. Therefore, baseline outcome benefit is zero.

Estimating project outcome:

In the project situation, the project outcome will be estimated based on the saved forest due to less fuel wood usage which will be surveyed during monitoring survey.

From the conducted survey usage rate with fire wood saving amount per household taken into account, then this value is converting into forest area by the national growing stock in the Cambodia with f_{NRB} factor. Therefore, baseline forest area usage calculated as:

$$SDG15 = N_{p,unit,,y} * U_{p,y} * (P_{b,wood,y} - P_{p,wood,y}) * f_{NRB} / \text{Growth stock}$$

Equation 19

$N_{p,unit,y}$	=	Monthly Average Number of total biodigesters installed in year y
$U_{p,y}$	=	Usage rate (%)
$fNRB_{i,y}$	=	Fraction of woody biomass used in the absence of the project activity in year y that can be established as non-renewable (95%)
Growth stock in forest	=	Growth stock in forest in Cambodia (162.15 tonne/ha)

Estimating net benefit:

For ex-ante estimation of SDG contribution, the Ex-ante value is estimated to be

Net benefit of SDG15= Project outcome of SDG15 – Baseline outcome of SDG15

Amount of forest area saved=

1,489 (Household Units) * 0.81(%) * (1.085 t - 0.142 t) *0.95(f_{NRB}) / 162.15 (t/ha)

=6.702 ha/year

B.6.2 Data and parameters fixed ex ante

Data/parameter ID	BGTA 1
Data/parameter	**Avoidance of double counting or double claiming among project participants**
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with other parties directly involved with the project or programme.
Source of data	Project developer and project technology producer is same with the project owner, NPB. Also implementers (BCAs) and retailers are contracted workers for NBP. Form03 and Form12 are signed before installment of biodigester which states the ownership rights and intention of selling the emission reductions resulting from the project activity directed with all parties.Each biodigester have unique design to differentiate from the others in the market, also each parameter have unique serial number which also saved in the project sale database.
Additional comment	Form03 and form 12 evidences submitted to the VVB.

Data/parameter ID	BGTA 2
Data/parameter	**Avoidance of double counting or double claiming with other mitigation actions**
Unit	NA
Description	Review and analysis of mitigation actions in other voluntary market or UNFCCC/compliance mechanisms.
Source of data	Gold Standard project GS751, which is also developed by the NBP shares the same technology and boundary. Double counting with GS751 is avoided by maintaining one database with both projects where all units installed after 30/06/2021 (thus starting on 01/07/2021) belong to GS11138 and all those built prior, belong to GS751.For animal farm biodigester (covered lagoon biodigester) a separated database is developed and manage by NBP.NBP signed declaration of No Double counting, where it states have not applied to other carbon scheme or voluntary or compliance standard proggrammes.
Additional comment	Data base and household inclusion for each project showed for the VVB during site visit, also screenshots of the data base are provided.

Data/parameter ID	BGTA 3

Data/parameter	**Regulatory framework for provision of animal waste management and thermal energy services**
Unit	NA
Description	Evidence that the project does not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply/devices or fuel supply or use
Source of data	Ministry of Agriculture Forestry and fisheries (MAFF) issues Prakas No549 on 12 Dec 2018 by mandating the treatment of animal waste (covering lagoon) at the farm, the application of biogas generating from the treatment system is still lacking. There is no more regulations or guidance determined in project geographical boundary of Cambodia.
Additional comment	Prakas No549 original version in Khmer Language and translated versions are provided to VVB.

Data/parameter ID	BGTA 4
Data/parameter	**Project technology description**
Unit	NA
Description	The detailed description of the project technology (including both biodigester and biogas stove) shall include as a minimum: - manufacturer name (if applicable), - product name (if applicable), - technology type, -capacity characteristics (e.g., volume of digester14), - continuous useful energy output demonstration, - rated thermal efficiency of biogas stove - Any performance certifications from national standards body or certification body recognised by national standards body also shall be provided.
Source of data	Designing of the technology is done by NBP, and constructed by the BCAs and masons. Detailed description provided in the PDD section A.3 above.
Additional comment	Technical drawings and details of the characteristics have been provided to VVB.

Data/parameter ID	BGTA 5
Data/parameter	**Expected technical life of project technology**
Unit	Operating hours (e.g., “5,500 hours”) or time period (e.g., “ten

	years")
Description	The expected technical life of an individual project technology shall be defined in the PDD.
Source of data	The designer, the manufacturer and the installer of the project is NBP. It is defined and planned to build more than 15 years of lifetime as stated in section C.1.2.
Additional comment	Technical drawings and details of the characteristics have been provided to VVB. Technical life time is longer than the crediting period.

Data/parameter ID	BGTA 6
Data/parameter	**Baseline scenario survey results**
Unit	NA
Description	Report of the results of the baseline scenario survey
Source of data	NBP Carbon Monitoring Survey (CMS) CP3 MP1 and NBP&ATEC's baseline survey(2018) for household biodigesters and NBP baseline survey for animal farms.(2023)
Additional comment	Baseline survey for both level of application, household and animal farms, have been provided to VVB.

SDG13

Data/parameter ID	BGTA 13 and BGTA15
Data/parameter	**$EF_{b,i,CO2}$ and $EF_{p,i,CO2}$**
Unit	tCO_2/TJ fuel
Description	CO_2 emission factor arising from use of fuels i in the baseline scenario (tCO2/TJ) and CO_2 emission factor arising from use of fuels i in the project scenario (tCO2/TJ)
Source of data	Wood and Charcoal Methodology default values are applied, for other fuel (LPG) 2019 refinement for 2006 IPCC Guidelines defaults, see chapter 2 Stationary Combustion: http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html
Value(s) applied	<table><tr><th>Fuel b-p</th><th>EF_{CO2,} (tCO2/TJ)</th></tr><tr><td>LPG</td><td>63.1</td></tr><tr><td>Charcoal</td><td>112.0</td></tr><tr><td>Firewood</td><td>112.0</td></tr></table>

Choice of data or Measurement methods and procedures	**N/A**
Purpose of data	Calculation of baseline and project emission
Additional comment	Charcoal applied the emission factor for charcoal fuel.

Data/parameter ID	BGTA 14 and BGTA16
Data/parameter	**$EF_{b,i,noCO2}$ and $EF_{p,i,non\text{-}CO2}$**
Unit	tCO_2/TJ fuel
Description	Non-CO_2 emission factor arising from use of fuels i in the baseline scenario (tCO2/TJ) and Non-CO_2 emission factor arising from use of fuels i in the project scenario (tCO2/TJ)
Source of data	Wood and Charcoal Methodology default values for AR5 GWP
Value(s) applied	**Fuel *b-p*** / **EF non-CO2, (tCO2e/TJ)**: Charcoal 9.46; Firewood 5.865
Choice of data or Measurement methods and procedures	For wood and charcoal, AR5 GWP values are used for combustion only.
Purpose of data	Calculation of baseline and project emission
Additional comment	Charcoal applied the emission factor for charcoal fuel. For other fuel (LPG) calculation is made by related parameters are based on IPCC default values and listed as fixed monitoring parameters as well calculation made with $(GWP_{CH4}*EF_{CH4})+(GWP_{N2O}*EF_{N2O})$ formula.

Value(s) applied:

Fuel *b-p*	**EF non-CO2, (tCO2e/TJ)**
Charcoal	9.46
Firewood	5.865

Data/parameter	**EF_{CH4}**
Unit	tCH_4/TJ fuel
Description	CH_4 emission factor arising from use of fuels in the baseline scenario and continued use of baseline fuels in the project scenario

Source of data	2019 refinement for 2006 IPCC Guidelines defaults see chapter 2 Stationary Combustion: http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html, table 2.9
Value(s) applied	<table><tr><th>Fuel *i*</th><th>EF_{CH4} (t/TJ)</th></tr><tr><td>LPG</td><td>0.001195</td></tr></table>
Choice of data or Measurement methods and procedures	IPCC default value
Purpose of data	Calculation of baseline and project scenario
Additional comment	This parameter is used to calculate non-CO_2 emissions factor for LPG . Multiplying this value by the GWP_{CH4} will be used to calculate.

Data/parameter	**$EF_{i,N2O}$**
Unit	tN_2O/TJ fuel
Description	N_2O emission factor arising from use of fuels in the baseline scenario and continued use of baseline fuels in the project scenario
Source of data	2019 refinement for 2006 IPCC Guidelines defaults, see chapter 2 Stationary Combustion, table 2.9 http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html
Value(s) applied	<table><tr><th>Fuel *i*</th><th>$EF_{i,N20}$ (t/TJ)</th></tr><tr><td>LPG</td><td>0.0021</td></tr></table>
Choice of data or Measurement methods and procedures	IPCC default value
Purpose of data	Calculation of baseline and project scenario
Additional comment	This parameter is used to calculate non-CO_2 emissions factor for LPG . Multiplying this value by the GWP_{N2O} will be used to calculate.

Data/parameter ID	BGTA 17 and BGTA18
Data/parameter	**$NCV_{b,I,fuel}$ and $NCV_{p,i,fuel}$**
Unit	TJ/ton
Description	Net calorific value of the fuel i that is substituted in baseline b and Net calorific value of the fuel i that is used in the project p
Source of data	For Charcoal and Wood Default values from methodology and for the other fuel (LPG) 2019 Refinement 2006 IPCC Guidelines defaults, see chapter 1 Energy table 1.2 http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html
Value(s) applied	Fuel *i* / $NCV_{b,ifuel,}$ / $NCV_{p,i,fuel}$ (TJ/ton): LPG 0.0473; Charcoal 0.0295; Firewood 0.0156
Choice of data or Measurement methods and procedures	N/A
Purpose of data	Calculation of baseline and project scenario
Additional comment	For wood and charcoal, methodology default emission factors are used. For the other fuel, IPCC value is used.

Fuel *i*	**$NCV_{b,ifuel,}$ / $NCV_{p,i,fuel}$ (TJ/ton)**
LPG	0.0473
Charcoal	0.0295
Firewood	0.0156

Data/parameter ID	BGTA 20
Data/parameter	**$P_{b,i,y}$**
Unit	tonnes/hh/yr
Description	Average yearly consumption of baseline fuel i per household before the start of the project activity or at the renewal of each crediting period, which is later (tonnes/household/year)
Source of data	Baseline Fuel Test (BFT) and CMS survey Ex-Ante source (BFT and CMS survey of NBP GS751 CP3 MP1&CP3MP2&CP3MP3)

Value(s) applied	<table><tr><th>Scenarios</th><th>Average per household ton/hh/yr</th></tr><tr><td>b1,wood</td><td>1.08593</td></tr><tr><td>b2,charcoal</td><td>0.01467</td></tr><tr><td>b3,lpg</td><td>0.01533</td></tr></table>
Measurement methods and procedures	$P_{b,i,y}$ is calculated based on the below formula. $P_{b,i,y} = B_{b,i,y} * B_{b,ratio}$
Purpose of data	Calculation of baseline emissions
Additional comment	Values are below the methodology threshold value and the cap values. Fixed for one crediting period. It is not practical to establish LPG consumption using the BFT for reasons of safety. Instead, households were be asked how often they buy a bottle and the bottle size will be recorded. Annual consumption was be calculated as: $Annual\ LPG\ consumption\ (\frac{kg}{year}) = \frac{Bottle\ size\ (kg)}{Usage\ period\ (days)} \times 365$ For all scenarios $B_{b,i,y}$, ratio of each scenario $B_{b,ratio}$ also determined by using survey methods to calculate weighted average fuel per household yearly

Data/parameter	$B_{b,i,y}$
Unit	tonnes/hh/yr
Description	Average yearly consumption of baseline fuel i per household for scenarios b before the start of the project activity or at the renewal of each crediting period, which is later (tonnes/household/year)
Source of data	BFT and CMS survey Ex-Ante source (BFT and CMS survey of NBP GS751 CP3 MP1 and CP3MP2)
Value(s) applied	<table><tr><th>Scenarios</th><th>Wood</th><th>Charcoal</th><th>LPG</th></tr></table>

	<table><tr><th></th><th>ton/hh/yr</th><th>ton/hh/yr</th><th>ton/hh/yr</th></tr><tr><td>b1,wood</td><td>1.14582</td><td>0.0</td><td>0.01305</td></tr><tr><td>b2,charcoal</td><td>0.02428</td><td>0.61142</td><td>0.02970</td></tr><tr><td>b3,lpg</td><td>0.31439</td><td>0.0</td><td>0.07885</td></tr></table>
Measurement methods and procedures	BFT is determined with a Kitchen Performance Test as per the requirements following 90/10 rule with 3 scenarios.
Purpose of data	Calculation of baseline emissions
Additional comment	for LPG scenario it is determined in CP3MP2, It is not practical to establish LPG consumption using the BFT for reasons of safety. Instead, households were be asked how often they buy a bottle and the bottle size will be recorded. Annual consumption was be calculated as: $$Annual\ LPG\ consumption\ (\frac{kg}{year}) = \frac{Bottle\ size\ (kg)}{Usage\ period\ (days)} \times 365$$ For all scenarios, ratio of usage also determined by using survey methods to calculate weighted average fuel consumptions $P_{b,i,y}$

Data/parameter	$B_{b,ratio}$
Unit	%
Description	Baseline fuel consumption scenario ratios
Source of data	CMS survey Ex-Ante source (CMS of GS751 CP3 MP3)
Value(s) applied	<table><tr><th>Scenarios</th><th>HH</th></tr><tr><th></th><th>%</th></tr><tr><td>b1,wood</td><td>93.9%</td></tr><tr><td>b2,charcoal</td><td>2.4%</td></tr><tr><td>b3,lpg</td><td>3.0%</td></tr></table>

Measurement methods and procedures	Determined by using survey methods
Purpose of data	Calculation of baseline emissions
Additional comment	Households will be asked which baseline scenario they fell into before receiving a biogas digester. The four baseline scenarios are defined by asking the households "what is your primary fuel source for cooking purposes?". The purpose of this question is to allocate a household to a baseline scenario, whereby 'primary' fuel consumption is defined as that fuel meeting more than 50% of their fuel needs for cooking purposes. The households that respond "firewood", for example, are subsequently allocated to the baseline scenario b1. This means that households in, for example, baseline scenario b1 may also use small amounts of fossil fuels (i.e. secondary and tertiary fuels) for cooking, which will be included in the baseline emissions.

Data/parameter ID	BGTA 7
Data/parameter	$\mathbf{GWP_{CH4}}$
Unit	tCO_2e per tCH_4
Description	Global Warming Potential (GWP) of methane
Source of data	AR5 IPCC
Value(s) applied	28
Choice of data or Measurement methods and procedures	N/A
Purpose of data	Calculation of baseline and project scenario
Additional comment	Shall be updated to any future COP/MOP decisions

Data/parameter	$\mathbf{GWP_{N2O}}$
Unit	tCO_2e per tN_2O

Description	Global Warming Potential (GWP) of nitrous oxide
Source of data	AR5 IPCC
Value(s) applied	265
Choice of data or Measurement methods and procedures	N/A
Purpose of data	Calculation of baseline and project scenario
Additional comment	Shall be updated to any future COP/MOP decisions

Data/parameter ID	BGTA 8
Data / Parameter	**$MS\%_{bl,j}$**
Unit	%
Description	Fraction of animal manure handled in baseline animal manure management system j. (%).
Source of data	2018 Baseline Survey (NBP_ATEC)
Value(s) applied	The next table shows the share in % of manure fed into the bio-digesters. (see table below)
Measurement methods and procedures	Method 2 – If animal manure is treated in different treatment systems manure weight delivered to each system shall be directly measured or alternatively manure volume can be measured together with the density determined from representative sample (90/10 precision). The quantity of animal manure from different farms and different animal types shall be recorded separately for cross-check. Recording of the baseline animal manure management system where

AWMS	Cow (other cattle)	Pig	Buffalo
Slurry	5.26%	31.74%	12.70%
Anaerobic Lagoon	0.25%	11.19%	3.03%
Dry lot	3.62%	10.13%	3.98%
Daily spread	7.71%	5.87%	7.33%
Solid storage	54.51%	36.17%	33.74%
Other	0.25%	3.41%	0.00%
Pasture	28.39%	1.49%	39.21%

	the animal manure would have been treated anaerobically is also required. Data will be collected according to the CMS sampling plan
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of contribution to SDG13
Additional comment	AWMS Calculation

Data/parameter ID	BGTA 10
Data/parameter	$VS_{Rate,LT}$
Unit	kg VS / (1000 kg animal mass) / day
Description	Default Volatile solids production/excretion rate per animal of livestock LT in year y.
Source of data	The country specific default not available, apply IPCC default. Volume 4 of the 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, chapter 10 (online: https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch10_Livestock.pdf)
Value(s) applied	**Animal** / **kg VS / (1000 kg animal mass) / day**: Pig: 9.8 Buffalo: 6.8 Cow: 13.5
Choice of data or Measurement methods and procedures	The calculation is a simple linear adjustment, so in the case of an animal that is 500 kg of weight, the VS emission rate will be half of the rate presented per 1000 kg live weight.
Purpose of data	Calculation of baseline scenario
Additional comment	IPCC Asia Mean Values are used, and calculation made with default TAM values proportional VS rates and nd_y (number of days treatment plant) to get $VS_{LT,y}$

Animal	kg VS / (1000 kg animal mass) / day
Pig	9.8
Buffalo	6.8
Cow	13.5

Data/parameter ID	BGTA 11

<table>
<tr><td>Data/parameter</td><td>Bo_{LT}</td></tr>
<tr><td>Unit</td><td>m^3 CH_4 /kg-dm</td></tr>
<tr><td>Description</td><td>Maximum methane production capacity for manure produced by livestock category T</td></tr>
<tr><td>Source of data</td><td>Country specific Bo values are not available. Volume 4 of the 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, chapter 10 (online: https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch10_Livestock.pdf)</td></tr>
<tr><td>Value(s) applied</td><td><table><tr><td>Animal LT</td><td>$Bo_{(LT)}$ m^3CH_4/kg-dm</td></tr><tr><td>Pig</td><td>0.29</td></tr><tr><td>Buffalo</td><td>0.10</td></tr><tr><td>Cattle</td><td>0.13</td></tr></table></td></tr>
<tr><td>Choice of data or Measurement methods and procedures</td><td>N/A</td></tr>
<tr><td>Purpose of data</td><td>Calculation of baseline and project scenario</td></tr>
<tr><td>Additional comment</td><td>N/A</td></tr>
</table>

Data/parameter ID	BGTA 12
Data/paramete r	$MCF_{j,k}$
Unit	%
Description	Annual Methane conversion factors for the baseline animal manure management system j by climate region k
Source of data	IPCC default values for the region Warm, Tropical Wet from volume 4 of the 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, chapter 10 (https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch10_Livestock.pdf)

<table>
<tr><td>Value(s) applied</td><td><table><tr><th>Anaerobic lagoon</th><th>Slurry</th><th>Dry lot</th><th>Daily spread</th><th>Solid storage</th><th>Other</th><th>Pasture</th></tr><tr><td>80%</td><td>74.22%</td><td>2%</td><td>1%</td><td>5%</td><td>1%</td><td>2%</td></tr></table></td></tr>
</table>

Choice of data or Measurement methods and procedures	The site annual average temperature is taken from official historical data (Cambodia 27.64 C °)[26] .As per requirement of the methodology and sourced from Table 10.17, Chapter 10, Volume 4 of the 2019 Refinement to 2006 IPCC Guidelines The IPCC is a standard, credible source of emissions factors.
Purpose of data	Calculation of baseline and project scenario from AWMS application
Additional comment	Cambodia located in Warm Tropical Wet for 10 months and in Warm Tropical Dry for 2 months based on the average monthly temperature defined in IPCC calculated in ER spreadsheet tab Input_DefaultValue B92:F102

Data/parameter ID	BGTA 41
Data / Parameter	$f_{NRB,i,y}$
Unit	%
Description	Fractional non-renewability status of woody biomass fuel during year y, in case the baseline fuel is biomass or charcoal.
Source of data	NBP 29 September 2021 Cambodia f_{NRB} assessment report
Value(s) applied	95
Measurement methods and procedures	Determined by following the CDM TOOL30, Calculation of the fraction of non-renewable biomass.
Monitoring frequency	Determined ex-ante and fixed for a given crediting period

[26] https://climateknowledgeportal.worldbank.org/country/cambodia/climate-data-historical

Purpose of data	Calculation of baseline and project emissions
Additional comment	

Data/parameter	$EF_{CO2,y}$
Unit	tCO_2/MWh
Description	CO_2 emission factor for grid electricity in year y
Source of data	Combined margin CO_2 emission factor published by Institute for Global Environmental Strategies (IGES) and National Council for Sustainable Development, Cambodia (2016)
Value(s) applied	0.3089
Choice of data or Measurement methods and procedures	The $EF_{grid,CM,y}$ is calculated using Operating margin CO_2 emission factor - $EF_{grid,OM,y}$ and Build margin CO_2 emission factor - $EF_{grid,BM,y}$ which were published by the IGES Cambodia as per "*Tool to calculate the emission factor for an electricity system*" – Version 04.0
Purpose of data	For calculation of $BE_{elec,y}$ Baseline CO_2 emission in the animal farms
Additional comment	Latest officially published data from 2016 $EF_{grid,CM,y}$

Data/parameter	$EF_{CO2,diesel,y}$
Unit	tCO_2/GJ
Description	Weighted average CO_2 emission factor of diesel fuel in year y
Source of data	IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in table 1.4 of Chapter1 of Vol. 2 (Energy) of the 2019 Refinement to the 2006 IPCC Guidelines on National GHG Inventories
Value(s) applied	0.0726
Choice of data or Measurement methods and procedures	No Values provided by the fuel supplier in invoices, according to tool03, IPCC default values used
Purpose of data	For calculation project emission from diesel usage in the animal farms

Additional comment	Any future revision of the IPCC Guidelines will be taken

Data/parameter	**$NCV_{diesel,y}$**
Unit	GJ/Kg
Description	Weighted average net calorific value of diesel fuel in year y
Source of data	2019 Refinement to the 2006 IPCC Guidelines defaults, see chapter 1 Energy table 1.2 http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html
Value(s) applied	0.0439
Choice of data or Measurement methods and procedures	No Values provided by the fuel supplier in invoices, according to tool regional default values used
Purpose of data	Calculation of project scenario from electricity application in the animal farms
Additional comment	Any future revision of the IPCC Guidelines will be taken

SDG 15

Data/parameter	**Growth stock in forest**
Unit	Tonne/hectare
Description	Growth stock in forest in Cambodia
Source of data	Cambodia growing stock: 94 m^3/ha (Global Forest Resources Assessment 2015, Page 79, Table 13 Growing stock in forest and other wooded land 2015)

	Converting factor from m3 of wood to tonne: 1.725 tonne/m3 (Chapter 3: GPG-LULUCF2003, sector Good Practice Guidance 2019 Refinement to the IPCC 2006, page 12)
Value(s) applied	162.15
Choice of data or Measurement methods and procedures	= Cambodia growth stock in forest (m3/ha)*Converting factor from m3 of wood to tonne of wood. = 94*1.725=162.15 tonne/hectare
Purpose of data	Calculation of project contribution for SDG 15
Additional comment	

B.6.3 Ex ante estimation of SDG Impact

>>The proposed approach for calculation of each SDG outcomes with detailed description described in above Section B.6.1 of this PDD. Please refer to the detail calculation in the ER spreadsheet, tab "SDG_Summary". Below in the table summary of results shown:

Item	Indicator	Baseline estimate	Project estimate	Net SDG outcomes
SDG 13	Amount of emission reduction in households (tCO_2e/yr)	5,032	1,588	3,444
	Amount of emission reduction in animal farms (tCO2e/yr)	29,046	10,772	18,274
SDG 1	Financial Savings from avoided biomass purchase (USD/year)	0	85,792	85,792
SDG 2	Total area on which bio-slurry is applied (ha/yr)	0	812	812
SDG 7	Household access to clean energy with the biodigesters sold under the project activity(#hh)	0	667	667
	Electricity consumption by users produced by the project Renewable energy (MWh/yr)	0	3,015	3,015
SDG 8	Number of jobs created by the project activity (#)	0	80	80

SDG 15	Forest area saved(ha/yr)	0	6.702	6.702

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 13 (The total amount of emission reduction (tCO2))

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
01.07.2021-30.06.2022	4,370	1,580	2,789
01.07.2022-30.06.2023	15,314	5,535	9,778
01.07.2023-30.06.2024	30,146	10,918	19,227
01.07.2024-30.06.2025	48,924	17,748	31,176
01.07.2025-30.06.2026	71,636	26,019	45,616
Total	108,586		
Total number of crediting years	5		
Annual average over the crediting period	21,717		

SDG 1.a (Financial savings from avoided biomass purchase (USD/year))

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
01.07.2021-30.06.2022	0	12,597	12,597
01.07.2022-30.06.2023	0	44,434	44,434
01.07.2023-30.06.2024	0	80,657	80,657
01.07.2024-30.06.2025	0	122,250	122,250
01.07.2025-30.06.2026	0	169,021	169,021
Total	428,958		
Total number of crediting years	5		
Annual average over the crediting period	85,792		

SDG 2 (Total area on which bio-slurry is applied)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
01.07.2021-30.06.2022	0	119	119
01.07.2022-30.06.2023	0	421	421
01.07.2023-30.06.2024	0	764	764
01.07.2024-30.06.2025	0	1,157	1,157
01.07.2025-30.06.2026	0	1,600	1,600
Total	4,061		
Total number of crediting years	5		
Annual average over the crediting period	812		

SDG 7 (Household access to clean energy with the biodigesters sold under the project activity)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
01.07.2021-30.06.2022	0	458	458
01.07.2022-30.06.2023	0	585	585
01.07.2023-30.06.2024	0	674	674
01.07.2024-30.06.2025	0	763	763
01.07.2025-30.06.2026	0	853	853
Total	3,333		
Total number of crediting years	5		
Annual average over the crediting period	667		

SDG 7 (Consumption of renewable energy-based electricity (MWh))

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT

01.07.2021-30.06.2022	0	377	377
01.07.2022-30.06.2023	0	1,319	1,319
01.07.2023-30.06.2024	0	2,638	2,638
01.07.2024-30.06.2025	0	4,334	4,334
01.07.2025-30.06.2026	0	6,407	6,407
Total	15,075		
Total number of crediting years	5		
Annual average over the crediting period	3,015		

SDG 8 (Number of jobs created)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
01.07.2021-30.06.2022	0	80	80
01.07.2022-30.06.2023	0	80	80
01.07.2023-30.06.2024	0	80	80
01.07.2024-30.06.2025	0	80	80
01.07.2025-30.06.2026	0	80	80
Total	400		
Total number of crediting years	5		
Annual average over the crediting period	80		

SDG 15 (Forest area saved ha/yr)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
01.07.2021-30.06.2022	0	0.984	0.984
01.07.2022-30.06.2023	0	3.471	3.471
01.07.2023-30.06.2024	0	6.301	6.301
01.07.2024-30.06.2025	0	9.551	9.551
01.07.2025-30.06.2026	0	13.204	13.204
Total	33.512		

Total number of crediting years	5
Annual average over the crediting period	6.702

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

Data/parameter ID	BGTA 24
Data/parameter	**Avoidance of double counting or double claiming among project technology end users**
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with project technology end users
Source of data	Form03, Form09 and Form12 are signed before installment of biodigester which states the ownership rights and intention of selling the emission reductions resulting from the project activity directed with all parties.
Monitoring frequency	Monitored whenever project technology is sold or otherwise disseminated
QA/QC procedures	Cross check using general internet search and search of public records of Gold Standard and other voluntary market and UNFCCC mechanisms
Additional comment	Form03 and form 12 evidences submitted to the VVB.

SDG 1&2&7&13&15

Data/parameter ID	BGTA 25
Data / Parameter	$\mathbf{U_{p,y}}$
Unit	%

Description	Average usage rate in project scenario p during year y
Source of data	Carbon Monitoring Survey (CMS)
Value(s) applied	Ex-ante estimation is based on figures, which are from NBP's CP3-MP4 GS751 project. <table><tr><th>Age Group</th><th>Usage</th></tr><tr><td>0.0 - 1.0</td><td>90.00%</td></tr><tr><td>1.0 - 2.0</td><td>90.00%</td></tr><tr><td>2.0 - 3.0</td><td>90.00%</td></tr><tr><td>3.0 - 4.0</td><td>61.29%</td></tr><tr><td>4.0 - 5.0</td><td>75.56%</td></tr><tr><td>Average</td><td>81.37%</td></tr></table>
Measurement methods and procedures	They will be updated by conduct the usage survey before the 1st verification of this PDD. (a)Survey of the users at randomly selected sample sites; where sampling is applied, the "Standard for sampling and surveys for CDM project activities and programme of activities" shall be used for determining the sample size to achieve 90/10 (for annual monitoring)
Monitoring frequency	Annual (survey of the users at randomly selected sample sites for household biodigester and all farms for cover lagoon biodigester)
QA/QC procedures	Compliance with the general requirements for sampling 9sectionn 4.4) and general requirement for QA/QC (Section 4.5)
Purpose of data	Calculation of baseline emissions, SDG1-2-7-13-15 contributions
Additional comment	

Data / Parameter	$N_{p,unit,y}$
Unit	#Units Number
Description	Monthly Average Number of total biodigesters installed in year y
Source of data	NBP Project Sale Database

Value(s) applied	<table><tr><th>$N_{p,unit-HH,y}$</th><th>$N_{p,unit-afarms,y}$</th></tr><tr><td>1,489</td><td>48</td></tr></table>
Measurement methods and procedures	100% of all plants are checked after completion of the construction by the PBP technician or by the BCAs. The main goal of NBP is to develop a market-based biogas programme and will gradually allow experienced BCAs to take over the commission check. The results of the commissioning check are recorded in Form 09, this is currently a physical document but in the coming years this may transit to using tablets. Only when Form 09 is approved by NBP, data is entered into the central database. All Form 03, and Form12 the hard copies, are stored at NBP.
Monitoring frequency	Annual (Census of users or survey of the users at randomly selected sample sites)
QA/QC procedures	Transparent data analysis and reporting. Hard copies of all construction contracts will be available and kept at NBP
Purpose of data	Calculation of baseline emissions, SDG1-2-7-13-15 contributions
Additional comment	This parameter is annually average of total installed biodigesters in each month, where the installation is determined monthly basis. It is calculated as cumulative value of monthly average of total biodigester number where the new sale data with 1 month delay

Data / Parameter	$N_{p,biodigester,y}$
Unit	#Units Number
Description	Number of biodigester sale in year y
Source of data	NBP Project Sale Database
Value(s) applied	<table><tr><th>$N_{p,biodigester-HH,y}$</th><th>$N_{p,biodigester-afarms,y}$</th></tr><tr><td>667</td><td>24</td></tr></table>
Measurement methods and procedures	100% of all plants are checked after completion of the construction by the PBP technician or by the BCAs. The main

	goal of NBP is to develop a market based biogas programme and will gradually allow experienced BCAs to take over the commission check. The results of the commissioning check are recorded in Form 09, this is currently a physical document but in the coming years this may transit to using tablets. Only when Form 09 is approved by NBP, data is entered into the central database. All Form 03,and Form12 the hard copies, are stored at NBP.
Monitoring frequency	Annual (Census of users or survey of the users at randomly selected sample sites)
QA/QC procedures	Transparent data analysis and reporting. Hard copies of all construction contracts will be available and kept at NBP
Purpose of data	Calculation of baseline emissions, SDG1-2-7-13-15 contributions
Additional comment	This parameter is annually average of sold biodigesters in year y .It is calculated with average of new sale data with 1 month delay, by taking enough time to for biogas production.

SDG1-13-15

Data/parameter ID	BGTA 42
Data/parameter	$P_{p,l,y}$
Unit	ton/hh/yr
Description	Average yearly consumption of project fuel I per household
Source of data	Carbon Monitoring Survey (CMS)
Value(s) applied	(see table below) These figures are based on the latest available data from NBP's carbon monitoring survey (CMS) for GS751 CP3-MP3.

Fuel *i*	**Average per household ton/hh/yr**
Firewood	0.14229
Charcoal	0.01052
LPG	0.01910

Choice of data or Measurement methods and procedures	For the LPG. It is not practical to establish LPG consumption using the PFT for reasons of safety. Instead, households were be asked how often they buy a bottle and the bottle size will be recorded. They will be updated by conduct the Project performance field tests before the 1st verification of this PDD.
Monitoring frequency	Updated for every 2 years or more frequently after the first verification The PFT values are valid for two years and may be applied for before or after period.
QA/QC procedures	Transparent data analysis and reporting The test, including LPG, achieved a level of precision/confidence of 90/27 and thereby meeting the requirements
Purpose of data	Calculation of project emissions
Additional comment	N/A

SDG 13

Data/parameter ID	BGTA 39
Data / Parameter	$LE_{p,y}$
Unit	tCO2e per year
Description	Leakage in project scenario p during year y
Source of data	Default value of the applied methodology is adopted
Value(s) applied	0.95 ER_{th}
Measurement methods and procedures	Updated data will be applied if there is any change in newer version of methodology
Monitoring frequency	default discount value of 0.95 applied to emission reductions
QA/QC procedures	Default value is applied and default discount value will be applied
Purpose of data	Calculation of project emissions
Additional comment	Physical leakage of the bio-digester

Data/parameter ID	BGTA 26
Data / Parameter	$N_{LT,y}$
Unit	Number
Description	Annual average number of animals of type LT in year y, in Households and animal farms
Source of data	Carbon Monitoring survey (CMS)
Value(s) applied	(see table below) Ex-ante numbers are get from NBP GS751 Carbon Monitoring survey(CMS) CP3 MP3 and NBP animal farm baseline-project monitoring survey .
Measurement methods and procedures	In the animal farm level, all the records of buy – sold pigs included if any deaths recorded by the farmers. Fattening pig farms with mother pigs, store all information about new borns. In the household level, Households give the information about their animal types and numbers during the carbon monitoring survey. Survey of the users at randomly selected sample sites will be used with photographic evidence where determining the sample size to achieve 90/10 rule.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting.
Purpose of data	Calculation of emission reductions
Additional comment	AWMS Calculation

Animal	N(LT),y HouseHolds	N(LT),y animal farms
LT	#	#
Cow (other cattle)	5.35	0
Pig	3.20	3437.5
Buffalo	0.83	0

Data/parameter ID	BGTA 27

Data / Parameter	**TAM_{LT}**
Unit	Kg/animal
Description	Typical animal mass for livestock LT at the project site.
Source of data	Default value depending on the productivity system. The relevant IPCC default TAM values as provided in Table 10A.5 as well as in the Annexes of Chapter 10 of chapter 'Emissions from Livestock and Manure Management' under the volume 'Agriculture, Forestry and other Land use' of the 2019 IPCC Guidelines for National Greenhouse Gas Inventories shall be selected accordingly.
Value(s) applied	IPCC Table 10A.5 -Asia Mean Values Other cattle:299 Buffalo:336 Breeding Swine: 122 Swine:58 Finishing:49
Measurement methods and procedures	-
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	Calculation of emission reductions
Additional comment	AWMS Calculation

Data/parameter ID	BGTA 28
Data / Parameter	nd_y
Unit	days
Description	Number of days the treatment plant was operational in year y
Source of data	Carbon Monitoring survey (CMS)

Value(s) applied	365
Measurement methods and procedures	Monitoring surveys
Monitoring frequency	Annually
QA/QC procedures	Determined using survey methods with the sample size to achieve 90/10 rule. Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	AWMS Calculation

<table>
<tr><td>Data / Parameter</td><td>$N_{da,y}$</td></tr>
<tr><td>Unit</td><td>-</td></tr>
<tr><td>Description</td><td>Number of days animal is alive in the farm in the year y</td></tr>
<tr><td>Source of data</td><td>Carbon Monitoring survey (CMS)</td></tr>
<tr><td>Value(s) applied</td><td>
<table>
<tr><th>Animal</th><th>$N_{da,HH,y}$</th><th>$N_{da,a\text{-}farm,y}$</th></tr>
<tr><td>Cow (other cattle)</td><td>365.00</td><td>0.00</td></tr>
<tr><td>Pig</td><td>365.00</td><td>300.00</td></tr>
<tr><td>Buffalo</td><td>365.00</td><td>0.00</td></tr>
</table>

Ex-ante estimations are adapted from NBP GS751 CP3MP3 and NBP animal farm baseline-project survey.</td></tr>
<tr><td>Measurement methods and procedures</td><td>In the households level, monitoring survey will be used to determine days of animal alive. To get annual number, Number of the animals are calculated with discount factor (for example, an household raise 10 fattening pigs per one round, two rounds per year, 150days per round, Total pig = 10pigs *2 round *150/365=8.22 pigs 78nnually. Thus, in household level it is 365 days.

For the animal farms level, animal inventory and sale data will be used to determine each batch of animal growing terms. In general each cycle takes 5 months, and 2 cycles per year with a month of gap. Thus, the operating days is 300 days.</td></tr>
<tr><td>Monitoring frequency</td><td>Annual, based on monitoring survey</td></tr>
</table>

QA/QC procedures	-Data will be collected according to the CMS sampling plan with the sample size to achieve 90/10 rule.
Purpose of data	Calculation of emission reductions
Additional comment	AWMS Calculation

Data/parameter ID	BGTA 30
Data / Parameter	$N_{p,y}$
Unit	-
Description	Annual average number of animals of type LT in year y.
Source of data	Carbon Monitoring Survey (CMS)
Value(s) applied	<table><tr><th>Animal</th><th>Np,y HouseHolds</th><th>Np,y animal farms</th></tr><tr><td>LT</td><td>#</td><td>#</td></tr><tr><td>Cow (other cattle)</td><td>5.35</td><td>0</td></tr><tr><td>Pig</td><td>3.20</td><td>3437.5</td></tr><tr><td>Buffalo</td><td>0.83</td><td>0</td></tr></table> Ex-ante estimations are adapted from NBP GS751 CP3MP3 and NBP animal farm baseline-project survey.
Measurement methods and procedures	Carbon Monitoring survey for both level and cross check with Animal Inventory data in animal farm level
Monitoring frequency	Annual, based on monitoring survey
QA/QC procedures	Data will be collected according to the CMS sampling plan with the sample size to achieve 90/10 rule.
Purpose of data	Calculation of emission reductions
Additional comment	This parameter is same with AWMS Calculation

Data/parameter ID	BGTA 31
Data / Parameter	$\mathbf{MS\%_{i,y}}$
Unit	%

Description	Fraction of manure handled in project animal manure management system i
Source of data	Carbon Monitoring Survey
Value(s) applied	The next table shows the share in % of manure fed into the bio-digesters. <table><tr><th>Animal</th><th>Biodigester (MS %) Households</th><th>Biodigester (MS %) animal farms</th></tr><tr><td>cow</td><td>58.37%</td><td>-</td></tr><tr><td>Pig</td><td>40.68%</td><td>100%</td></tr><tr><td>Buffalo</td><td>50.79%</td><td>-</td></tr></table>For ex-ante estimation, the values are applied following the latest survey result GS751 CP3 MR3 which was also developed by the PP. And developed baseline survey in animal farms level.
Measurement methods and procedures	Method 2 – If animal manure is treated in different treatment systems manure weight delivered to each system shall be directly measured or alternatively manure volume can be measured together with the density determined from representative sample (90/10 precision). The quantity of animal manure from different farms and different animal types shall be recorded separately for cross-check. Recording of the baseline animal manure management system where the animal manure would have been treated anaerobically is also required. Data will be collected according to the Carbon Monitoring Survey (CMS) sampling plan.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of contribution to SDG13
Additional comment	AWMS Project emission Calculation

Data/parameter ID	BGTA 38
Data / Parameter	$N_{b,p,y}$
Unit	days
Description	Number of project technology-days included in the project database for each project scenario in year y

Source of data	Calculated from the project database
Value(s) applied	365
Measurement methods and procedures	Database will be used to determine project technology day.Biodigester Project sale database is generated on monthly storage basis. Depending on that ER calculation is made monthly basis where the number of biodigester taken into account individually for each month.
Monitoring frequency	Calculated Annually
QA/QC procedures	Cross check with usage survey.
Purpose of data	Baseline emission and project emission Calculation
Additional comment	For simplify the ER calculation per unit it is taken as whole year. Sale database is used to determine number credited units and the date they commissioned. Depending on that monthly ER calculations are made by taking usage rate into account.

SDG 7 & 13

Data / Parameter	$FC_{diesel,generator,y}$
Unit	Liters (lt/yr)
Description	Quantity of diesel fuel combusted in generator during the year y
Source of data	Carbon Monitoring Survey (CMS)
Value(s) applied	10800 lt/yr Ex-ante estimation is from Animal Farm Baseline-Project Survey
Measurement methods and procedures	Determined using survey methods
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting. The data will be analysed in the monitoring report and raw data will be available on request to the VVB.

Purpose of data	To estimate SDG13 contribution and calculate replaced grid electricity consumption
Additional comment	To make unit conversion, density of the diesel fuel from CDM methodology AMS.I-B,V.13.0, default value 0.8439 kg/lt is used to convert into kg/yr.

Data / Parameter	$EG_{BL,y}$
Unit	MWh/year
Description	Quantity of net electricity displaced in year y
Source of data	Electricity Bills (Survey input)
Value(s) applied	62.8 MWh
Measurement methods and procedures	Calculated from the input in the survey for baseline and project scenario of grid electricity bills which indicates consumptions.
Monitoring frequency	Annually
QA/QC procedures	According to AMS-I.F
Purpose of data	For baseline emission calculation.
Additional comment	N/A

SDG1

Data / Parameter	Biomass Fuel Price
Unit	$USD/tonne
Description	Annual Average Fuel Price for biomass
Source of data	Carbon Monitoring Survey (CMS)
Value(s) applied	75$/tonne Ex-ante estimation from HSE Monitoring Survey 2021
Measurement methods and procedures	Determined using survey methods
Monitoring frequency	Annually

QA/QC procedures	Transparent data analysis and reporting. The data will be analysed in the monitoring report and raw data will be available on request to the VVB.
Purpose of data	To estimate SDG1 contribution
Additional comment	

SDG2

Data / Parameter	$\%BS_y$
Unit	percentage
Description	Share of households that uses bio-slurry for crop production in year y
Source of data	Carbon monitoring Survey (CMS)
Value(s) applied	96.97% Ex-ante estimation, GS 751 CP3 MP3
Measurement methods and procedures	Determined using survey methods
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting. The data will be analysed in the monitoring report and raw data will be available on request to the VVB.
Purpose of data	To estimate SDG2 contribution
Additional comment	

Data / Parameter	Abs,h,y
Unit	Hectare(ha)
Description	Average area per household on which bio-slurry is applied in year y
Source of data	Carbon monitoring Survey (CMS)
Value(s) applied	0.69

	Ex-ante estimation, GS 751 CP3 MP3
Measurement methods and procedures	Determined using survey methods
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	To estimate SDG2 contribution
Additional comment	

SDG8

Data / Parameter	Number of jobs created by project activity
Unit	-
Description	Number of created jobs by project activity
Source of data	Annual Reports
Value(s) applied	80
Measurement methods and procedures	Please refer to section B.6.1 for the detail calculation method
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting. The data will be analysed in the monitoring report and raw data will be available on request to the VVB.
Purpose of data	To estimate SDG8 contribution
Additional comment	

B.7.2 Sampling plan

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For household biodigester, per registered methodology, two surveys and two tests need to be conducted. They are Usage Survey (US), Carbon Monitoring Survey (CMS), Project Performance Field test (PFT) and Baseline Performance Field Test (BFT). The sample frame of the survey and field test are shown in Figure 7, except the BFT in which sample frame are those who don't own biodigester, the others are those who own biodigester.

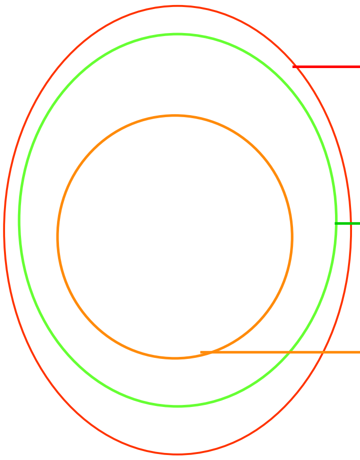

Figure 7. Sample frame of US, CMS and PFT for household biodigester activities

For farm biodigester (covered lagoon biodigester), 100% of the registered farm will be selected for the survey.

1. Sample design of US, CMS Survey

#	Item	Description
1	Objectives and Reliability Requirements	The objective is to obtain unbiased and reliable estimates of the monitoring parameters at a confidence / precision level of 90/10 (for annual monitoring) or 95/10 (for biennial monitoring).
2	Target population, Sampling method and sampling frame	**Household biodigester** Households that have installed an NBP certified biodigester are included for usage survey sampling frame and users who still using biodigesters are included in the sampling frame of CMS survey. The selection of provinces should be done based on both the Cambodia region classification (Plateau/mountains, Tonle Sap lake Zone, Plain Zone and Costal Zone) and the top sale province of each zone. This is done to reduce the burden on the finance for this survey without compromising its representativeness. The number of villages per selected provinces is calculated proportionally to its sale share of the provinces.
3	Sampling method and sampling frame	The sample frame for Usage survey: are all the end users (for both active user (still use biodigester) and no-active users (stop using biodigester) registered under the project as shown in Figure 7. They are classified into three levels: 1st level is for Usage survey samples, 2nd for CMS (users who still use biodigester), 3rd for PFT (users still use biodigesters).

		First the US is conducted, if the user is an active user, then CMS will be conducted otherwise move to next household. When a user joins the US and CMS, he/she will be asked to join the PFT *(see detail in below section, D. baseline and project fuel test)*.
4	Sample size	Sample size needs to be determined on a stratum level to ensure that the precision and reliability requirements are met throughout the data set. The size of the sample for each sampling frame is determined by the requirement to achieve the 90/10 or 95/10 confidence/precision for the estimation of the proportion or mean value of the parameter investigated. -**For CMS survey** where the number of animal will be collected the sample size will follow equation[27] $$c \geq \frac{\left(\frac{SD_B}{Clustermean}\right)^2 \times \left(\frac{M}{M-1}\right) + \left(\frac{1}{u}\right) \times \left(\frac{SD_W}{Overallmean}\right)^2 \left(\frac{\overline{N}-u}{\overline{N}-1}\right)}{\frac{0.1^2}{z^2} + \frac{1}{M-1}\left(\frac{SD_B}{Clustermean}\right)^2}$$ Where: c : Number of groups that should be sampled M : Total number of groups in the population u : Number of units to be sampled within each gro (pre-specified as 10 households) $\overline{N}$: Average number of units per group z : 1.645 for the 90% confidence and 1.96 for the confidence 0.1 : Represents the 10% relative precision The example for calculation method could be found in Example 8 – Multi-stage sampling, section 2.1.11. from page 41 of the Guideline “Sampling and Surveys for CDM Project Activities and Programmes of Activities”, Version 04.0

[27] Equation 33, example 2.1.11, page 44 of the Guideline “Sampling and Surveys for CDM Project Activities and Programmes of Activities”, Version 04.0

For usage survey: The sample size will be calculated based on the following equation[28]

$$c \geq \frac{\frac{SD_B^2}{\bar{p}^2} \times \frac{M}{M-1} + \frac{1}{\bar{u}} \times \frac{SD_w^2}{\bar{p}^2} \times \frac{(\bar{N}-\bar{u})}{(\bar{N}-1)}}{\frac{0.1^2}{z^2} + \frac{1}{M-1}\frac{SD_B^2}{\bar{p}^2}}$$

Where:

C	:	Number of groups that should be sampled
M	:	Total number of groups in the population
$\bar{u}$	:	Number of units to be sampled within each gr (pre-specified as 10 households)
$\bar{N}$	:	Average number of units per group
SD_B^2	:	Unit variance (variance between villages)
SD_W^2	:	Average of the group variances (Average with village variation)
$\bar{p}$	:	Overall proportion
z	:	1.645 for the 90% confidence and 1.96 for th confidence
0.1	:	Represents the 10% relative precision

The example for calculation method could be found in Example 4 – Multi-stage sampling, section 2.1.5 from page 33 of the Guideline “Sampling and Surveys for CDM Project Activities and Programmes of Activities”, Version 04.0.

[28] Equation 16, example 2.1.5, page 34 of the Guideline “Sampling and Surveys for CDM Project Activities and Programmes of Activities”, Version 04.0

2. Data

#	**Item**	**Description**
i	Field measurements	The survey will consist of household visit in random selected end-users with the objective to collect reliable and unbiased data. The data will be collected using interview methods, the interviewee will be either the head of the household or the wife of the head of household.
Ii	Quality Assurance/Quality Control	Several mechanisms will be put into place to avoid non-sampling errors (bias) and to obtain reliable date for each parameters: • **Good Questionnaire Design and piloting** The survey questionnaire will be developed and tested under real life conditions (pilot testing: taken to the field and tested with farmers as interviewees). The outcome of that testing will result in an improved questionnaire and will only be used after approval of NBP • **Field survey quality control** Experience enumerators will be selected and trained. The survey will be done using digital collecting tool (phone/table) and 100% of questionnaire will be checked by field supervisors/manager.
Iii	Procedures for Administering Data Collection and Minimizing Non-Sampling Errors	The survey team will Interview a ”Ando' selected household and answers will be recorded in a questionnaire, in case of non-response the surveyor will proceed to the next household. The surveyor will document the out of population cases, refusals and other sources of non-response. Also, the surveyor will only interview informed interviewees, i.e. interviewees with knowledge on cooking and the biogas plant. The original questionnaire used will be made available for inspection by VVB.

3. Implementation

#	**Item**	**Description**
i	**Implementation**	The US and CMS will be executed annually. The data collection will be executed by an independent entity which is selected on a number of criteria (experience, legal status of the company, quality of their proposal). Persons involved will have the following qualifications and experience: • Surveyors: Person that is trained by the survey team leader • Survey team leader: Experience person and has been involved in at least 2 other surveys • Reporting and PDD updating: Person that is involved in at least one verification or in large related surveys

ii	**Data storage arrangement:**	All data obtained from the US and CMS will be stored in a database, which will contain the data of the sampled households for each monitoring interval: 1. Location of each biogas plant surveyed; 2. Name of each biogas plant owner; 3. Unique code of each surveyed biogas plant; 4. Size of each surveyed biogas plant; 5. Type of biogas plant; 6. Name and ID of mason that built the biogas plant; 7. Number of animals (Pig, buffalo, cattle and dairy, Cow); 8. Fuel consumption (kg/year) of surveyed households (; 9. Date of commissioning for each plant;

D. Baseline and Project Fuel test

The baseline performance field test (BFT) and the project performance field test (PFT) measure real, observed technology performance in the field. Consumption is measured with a representative sample of end users under each defined baseline scenario (in the absence of the project technology) and project scenario.

The BFT/PFT is executed according to this protocol:

- The minimum recommended test period is 3 days as per methodology. During GS751 CPII and MPI of CPIII NBP however, NBP was given approval to reduce this period to 1 day as most households do not use fuels anymore other than biogas. Therefore, the test period shall be 1 days[29]
- Cooking practices includes will be fuels used for human food cooking and boiling water
- Cooking practices shall be during 'normal days'. Normal days are defined as periods without extra eaters. Depending on the family, this excludes days like festivals or holidays or weekend days. The MC can take place in the weekend if it can be proven that fuel use is not higher during these days (i.e. the same number of people eat meals as during the week).
- Households are instructed that they cook normally during the test. The aim is to capture their usual behaviour in the kitchen, as if no tests were happening
- To conduct the tests, ensured is that the cook uses fuel only from a designated stock which is pre-weighed.
- The number of person-meals will also be recorded in the following categories: Child 0-14 years, Female over 14 years, male 15-59 and male over 59 years old.

[29] *A MC of 1 days (24 hour) is allowed by the GS during CPI*

- Fuel will not be provided to the households, but they will be assisted with gathering enough wood if necessary. The reason being is that the main fuel, wood, is not purchased but collected.

The PFT design is depicted in the table below:

Table 6: PFT and BFT survey design

Item	Conversion in PFT	Conversion in BFT
Sampling objective:	The objective of the sampling effort is to obtain reliable fuel use data of project households	Idem but of baseline households
Field Measurement Objectives and Data to be collected:	The survey will consist of a 24 hour measurement campaign	idem
Target Population and Sampling Frame:	The sampling frame will be drawn from the project database	Households are those with the technical potential for biogas and with the same socio-economic and cultural practices as the PFT households and similar stove and fuel usage as collected by the baseline survey
Sampling method (approach):	Cluster random sampling using the CMS sampling frame with a minimum sample of 30 households	Neighbouring households to the PFT households
Implementation:	Biennial	Once in CPI and to be executed for MPI
Desired Precision/Expected Variance and Sample Size.	90/10 rule of the applied methodology or the lower bound of the one-sided 90% confidence interval in case 90/10 is not achieved	
Procedures for Administering Data Collection and Minimizing Non-Sampling Errors:	The test will be executed by a third party and 10% of the households will be double checked either by visits or telephone calls	

B.7.3 Other elements of monitoring plan

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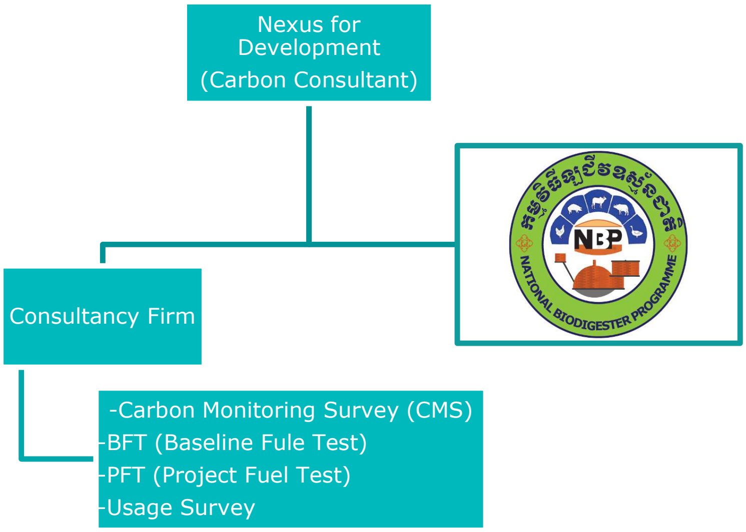

Figure 8 Organization Structure of Monitoring Team

Total sales or dissemination record:

NBP, created the database with the complete sale records form 09 for household biodigester and form12 for covered lagoon biodigester to collect the below basic information of each constructed biodigesters and maintain them in the database. The number of biodigester will be taken as previous month from the sale database to allow for a 1-month period for starting up the biodigester. This is conservative because in most cases within 2 weeks biogas is being produced. In the database following information are saved for each user:

- Date of installation
- Geographic area of sale
- Model/type of project technology sold
- Quantity of project technologies sold
- Name and telephone number (if available), and address and/or GPS coordinates
- Mode of use

Project Database:

The project database lists all the project technology units(biodigesters) for household and animal farm level that have been installed, sold or distributed by the project and have not surpassed their technical life. In the database all the information related to biodigester and user can be found. The number of biodigester will be taken as previous month from the sale database to allow for a 1-month period for starting up the biodigester. This is conservative because in most cases within 2 weeks biogas is being produced.

Quality control on plants in operation and under construction

Quality control on plants in operation and under construction is a key aspect of quality enforcement and the long-term success of the programme. The controls are conducted by supervisors of the Provincial Biodigester Programme (PBP) or in the future by experienced BCAs with regular assistance from the National Programme Office technicians. The latter also conduct quality control on the quality control performed by the PBPs on a random sample basis.

A PBP supervisor visits upon completion each biodigesters. The PBP supervisors monitor if the plant is completed according to standardized specifications. If the final quality of the plant is considered acceptable, the plant is officially handed over to the farmer and a plant completion report is filled and signed by both parties. With a copy of the report the farmer can claim the subsidy on the investment. All plant completion reports are entered into the programme database located at the department of agriculture in Phnom Penh.

Quality control on masons and BCAs

BCAs and mason teams who corporate with the PBP and benefit from the subsidy scheme, are required to seek recognition from NBP and the PBP office. Such recognition is subject to a series of strict conditions such as:

- approval of standard design and sizes of biodigesters;
- trained, certified and registered masons for the construction of biodigesters;
- construction of biodigesters on the basis of detailed quality standards;
- provision of PBP approved quality biodigester appliances (pipes, valve, stove, water trap, lamp);
- provision of proper user training and provision of a user instruction manual;
- provision of one year guarantee on appliances and two years guarantee on the civil structure of the biodigester, including an annual maintenance visit during the guarantee period;
- timely visit of a technician to the biodigester in case of a complaint from the user;
- Proper administration.

These conditions are put down in an agreement between the PBPs and the biodigester construction companies.

After sale services

After sales service is an integral part of the product delivered by individual masons and/or BCA. The after sales service include proper instruction of the user on the operation of the plant and maintenance as well as a 1year guarantee on appliances and 2 years on the civil structure of the plant. The guarantee provision includes at least 2 visits with a 1year interval, starting 3 months after the completion of the plant and a final visit one week before the end of warranty period. These visits have to be recorded on the backside of the guarantee certificate which is issued to the owner.

User trainings
The instruction of the user will include the following aspects of plant operation and maintenance:

- proper feeding of the plant;
- proper use of biodigester;
- regular simple maintenance like cleaning of the burner, changing the mantle of the lamp and the use of the water trap;
- proper use of the plant effluent;
- Cooking habits and cooking environment.

Avoidance of double counting

For domestic biodigester: Every plant is upon completion but before commissioning inspected by a PBP supervisor or BCA director. This is a structured inspection with the use of a Completion Report Form (form 09). This is also the moment when the plant is handed over from the company to the owner. The PBP supervisor or BCA director, the concerned mason and the plant owner will go through this inspection together and if they agree, the plant including the piping and the appliances installation, are completed, each one of them will sign the form while the PBP director will also stamp it.

Of this form the first page goes to the owner, the second to the NBP national office and a third one remains with the PBP.

For covered lagoon biodigester: Baseline and usage form (form 12) will be used to record all the information of the farm with plant code, biodigester plant size, address of the farm and its detail technical specification of the installed biogas generator.

NBP signed the declaration of no double counting, provided to VVB. According to applied Gold Standard methodology, Avoidance of double counting or double claiming among project participants(BGTA1) listed above in the section B.6.2 where it states that the project owner, technology producer and implementer is all done by NBC or its actors(PBp,BCA and masons). Avoidance of double counting or double claiming among project technology end users (BGTA24) listed above in the section B.7.1 and will be monitor with the signed forms. Furthermore, the following form are used for collecting user information, quality control of the constructed plant, end of warranty and aftersales services

Form no:	Type of biodigester	Activity	By whom:
Form 03	Domestic biodigester	Construction contract between client and BCA with a copy to PBP/NBP.	Client and BCA.
Form 06	Domestic biodigester	Quality control report. Quality control during the construction of a random sample of about 40% of all plants which are checked during the construction on quality standards. The forms are entered in the NBP Database and feedback is given to the PBPs and companies. Copies of the quality forms are kept at the PBP and the NBP office.	BCA conduct the quality control on their mason's work by filing the form and send to PBP/NBP. Data entry by NBP data typist, checked by NBP Information System Manager.
Form 09	Domestic biodigester	Completion Report. All plants, after completion but before commissioning, are checked by a PBP supervisor and client. If found that the plant is completed and up to quality standards, the plant is handed over to the owner for use. The owner can claim subsidy at a bank with his copy of the report. The data are entered at the NBP Database and copies are kept at the PBP and NBP office. Data are used for progress reports and subsidy transfer control.	PBP technical supervisor Data entry by NBP data typist, checked by NBP Information System Manager.
Form 10	Domestic biodigester	End warranty report: All plants at the end of two-year warranty, the survey will be conducted by mason for checking the final situation of the plant. All the reparation work will be done at this stage if needed. The summary report are kept and NBP office and entered into NBP Database while the full original copies are kept at the PBP.	BCA do the visit and fill in the form using Kobo toolbox, an online survey platform, checked by NBP technical unit team.
ASS-Form	Domestic biodigester	After Sale Services (ASS) form is used to monitor the biodigester performance in both quality of the biodigester plant and its usage. The forms are entered into the Dbase by NBP supervisor or the BCA who has done the check as part of the quality control or ASS respectively.	NBP technical supervisor and BCA do the check or visit and fill in the form using Kobo toolbox, an online survey platform, checked by NBP technical unit team.

Form 12	Covered lagoon biodigester	Baseline and usage form for covered lagoon biodigester.	PBP technical supervisor conduct the survey, fill in the form using Kobo toolbox. NBP technical unit conduct the quality control.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

>>

01/07/2021. This starting date is the date of implementation of the first biodigester under the project as shown in its contract agreement between the farmer and the biodigester construction agent (BCA).

C.1.2 Expected operational lifetime of project

>>

More than 15 years[30].

C.2. Crediting period of project

C.2.1 Start date of crediting period

>>

01/07/2021 which is the start date of the proposed project or 2 years prior to the date of Project Design certification whichever is later.

C.2.2 Total length of crediting period

>>5 years, twice renewable

[30] This lifetime is well observed in the other NBP project GS751 where the 16-year biodigester plant is still in operation, 23%, as reported in the usage rate of 16- age group.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is[97] summarized below.

N/A no issues identified that require monitoring

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1–- Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	NBP is meeting the foundational gender sensitive minimum standards[31]. For example, NBP has a gender sensitive design and ensures that both women and men are involved during trainings and village group meetings. For example, trainings are generally organized at a moment that is convenient for women and not interfering with their other (domestic) chores. NBP have women employees and also during the local stakeholder meeting, women contributors also attended the meeting.

[31] See page 10 paragaph 15 of https://www.goldstandard.org/sites/default/files/documents/gs_gender_policy-2.pdf

Question 2–- Explain how the project aligns with existing country policies, strategies and best practices	Key national policies are: • Neary Rattanak IV – five year strategic plan for Gender Equality and Women's empowerment 2014-2018 of the Ministry of Women's affairs (MOWA)[32] • MAFF Gender Mainstreaming Policy and Strategic Framework in Agriculture 2016-2020[33] The latter document is the most relevant, and addresses issues like time / labour to collect firewood limits women's capacity for higher value tasks which limits nutrition, health, education and life opportunities, limited access to energy that makes women's lives even more difficult. NBP have women employees, where they pay regard for the all issues.
Question 3–- Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, Gender is adequately addressed in the Safeguarding principles assessment.
Question 4–- Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, no particular gender challenges are expected

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

[32] *Available online: http://www.mowa.gov.kh/inc/uploads/2018/01/MoWA-Neary-Rattanak-IV-2014-2018-EN.pdf*

[33] MAFF*: Gender mainstreaming policy and strategic framework in agriculture 2016-2020*

E.1 Summary of stakeholder mitigation measures

Based on the physical workshop, two questions are relevant and are taken into consideration for this project. For domestic biodigester, NBP has already the list of BCA for each target province and NBP has also contracted with the local workshop to provide a quality biogas appliances including main gas pipe, biogas tap and stove to the users. For covered lagoon biodigester, NBP will make a list of good biodigester contractors and a list of generator installers. Regarding subsidy, NBP provided 150 USD for domestic biodigester but for cover lagoon biodigester, NBP may not have a direct subsidy at this moment but will provide assistance for recommending good biodigester installer and good generator installer.

Gender of Stakeholder	Stakeholder comment	Was comment taken into account (Yes/ No)?	Explanation/ Justification (Why? How?)
M	Did NBP work with the generator seller to find out what kind of generator for pumping water can be used with biodigester? high quality? long lasting used with the biodigester? Does the NBP have a service provider to repair or support?	Yes	NBP has conducted the research and development tested generators for 5.5 horsepower and 2.5 horsepower connected to household biodigesters (4m3 and 6m3). As a result, a generator connected biodigester can be used for between 15 minutes to 30 minutes, but if farmers use more than 30 minutes, the biodigester may be lacked of biogas for cooking. For covered lagoon biodigester, NBP will work with generator suppliers and biodigester installer to install a

			suitable size of that farm and good quality of generator. The list of service providers will be made available at NBP and PBP. The interested farm can contact them for further information.
M	Q3. How much subsidies does NBP provide? Because today the consume price of electricity is expensive during the day time, while the consume price of electricity can be cheaper if used from 7 pm to 4 am	Yes	NBP provides a subsidy of $ 150 for biodigester size from 3m^3 to 15m^3, but NBP has not yet considered the subsidy for covered large lagoon. NBP and the General Directorate of Animal Health and Production collaborates to provide an assessment of this case. Based on experience in other country like Vietnam, the government did not support any subsidy for covered large lagoon to farms because the covered large lagoons biodigester are run based on a commercial base. Furthermore, based on our field visit in Kampong Spue, the farm owner request if NBP could coordinate or provide technical support especially finding good technology providers (biodigester technician and biogas generator provider) because it is difficult to find good

			technician/biogas generator's providers especially when the generator is broken. Thus, at first step NBP wants the farm owners to invest by themselves, while NBP will coordinate in supporting them for finding good technology providers. Furthermore, any additional support to install monitoring equipment for carbon monitoring survey will be considered in a case by case during the project implementation.
M	Does the NBP have a service provider to repair or support?	Yes	The list of service providers will be made available at NBP and PBP. The interested farm can contact them for further information.

E.2 Final continuous input / grievance mechanism

METHOD	INCLUDE ALL DETAILS OF CHOSEN METHOD (S) SO THAT THEY MAY BE UNDERSTOOD AND, WHERE RELEVANT, USED BY READERS.
Continuous Input / Grievance Expression Process Book (mandatory)	The Expression Process book will be kept at NBP where all the complaints and grievances will be recorded or filed.
GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	Hotline: +855 (0)78 72 41 11

Internet/email access (optional)	nimol@nbp.org.kh; nimol.imm@gmail.com
Other	Facebook: https://www.facebook.com/nbp.org.kh/ Telegram: https://t.me/+Qy9FYHdlfthiNzc9

APPENDIX 1–- SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in	**No**	Cambodia has signed and ratified the "International Convention Economic, Social and Cultural Rights" and the "International Covenant on Civil and Political Rights"[34]. The project respects human rights, including dignity, cultural property and uniqueness of indigenous people. The installation of	N/A

[34] *https://tbinternet.ohchr.org/_layouts/TreatyBodyExternal/Treaty.aspx?CountryID=29&Lang=EN*

violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion		biodigesters relies on individual households voluntarily investing. The voluntary nature of this purchase will ensure that the individual dignity, cultural property and uniqueness of indigenous peoples are respected. Thus, the project is not complicit in Human Rights abuses . Cambodia ratified the ILO Convention C087 (Freedom of Association and Protection of the Right to Organise Convention, 1948) and C098 (Right to Organise and Collective Bargaining Convention, 1949)[35] All staffs are voluntary working under NBP and are free to form association and provide feedback. Employment at NBP, PBPO or franchised enterprises, BCAs, do not discriminate against individuals and employment of staff is not based on gender, race, religion, sexual orientation or on any other basis.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to	No	1. No, the project does not affect control of resources, entitlements and benefits as, on the contrary, it brings benefits on time and resources savings which are mainly accrued to women 2. Project does not adversely affect men and women in marginalised or vulnerable communities	N/A

[35] http://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:11200:0::NO::P11200_COUNTRY_ID:103055

adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of		1. No, on the contrary as per item 2 above. It frees up time. NBP is open to all but there are certain societal barriers that prevent women to attend certain meetings due to household chores. Therefore, users' trainings and small group meetings are organized at a time convenient for women, around 8:30-9:30. In fact, more women are reached than man, i.e in 2020 approximate 8,199 potential farmers of which 52% female attended the pre-construction trainings and 92 user training courses were conducted with total 1354 participants (734 women) [36] 3. The Project has taken into account gender roles and the abilities of women or men to benefit from the Project's activities. The Project is demand driven and any minority, if they have the ability to pay and enough manure, can invest. 4. On the contrary, it saves women a lot of time by no having to collect wood, faster and more convenient cooking and less cleaning of pots and pans as not soot is produced. 5. No, most of the benefits are accrued to women, see item 5 and the project does therefore not deepen discrimination 6. The Project does not limit women's ability to use, develop and protect natural resources as it would help women eliminate the need to collect firewood and therefore it protects natural resources. Biodigesters are a safe and established technology in Cambodia, so no risks are anticipated regardless of gender, women, girls or boys.	

[36] NBP 2020 Annual Report

opinions and recommendations of an Expert Stakeholder(s)		4.No need for expert stakeholder opinion or recommendation, since there is no challenges occurred in Gender and it is adequately addressed in the assessment.	
Principle 3. Community Health, Safety and Working Conditions			
The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community no	**No**	The Cambodian Constitution provides Cambodians with a range of rights and obligations such as Articles 228-232 of the Labour Law[37] 1.Biodigesters hygienically treat waste and eliminate household air pollution. Health consequently will be improved 2.All masons are trained and certified on safe working conditions. NBP obliges mason to buy an accident insurance which is the most dangerous activity.	N/A
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites,	**No**	Cambodia has signed and ratified the "Convention for the Safeguarding of the Intangible Cultural Heritage, 2003"[38]	N/A

[37] http://www.cambodiainvestment.gov.kh/the-labor-law-of-cambodia_970313.html

[38] http://portal.unesco.org/en/ev.php-URL_ID=33391&URL_DO=DO_TOPIC&URL_SECTION=201.html

structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture? >>		1. The project area are most provinces in Cambodia, but the site of the digesters is located at backyard of households farms and vicinity of animal farms generally next to animal barns without any objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture on a voluntary basis and does not involve and is not complicit in the alteration, damage or removal of any cultural heritage.	
Expert stakeholder Opinion for Principle 4.1: - No risks, the biodigesters are constructed and installed in the garden of the farmer, taking a plot as small as 5 by 4 meters for domestic biodigester and for medium and large scale biodigester, they are built within the premises of existing animal farms (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia). - Agree to the above statement. No risk associated with the construction and installation of biodigesters in the former's garden and animal farms. However, during the excavation for construction, in case artifacts or other similar objects found the construction work should be cease and inform related authority/institutions (opinion from Mr. Lytour Lor, director, Biogas Technology and Information Center (BTIC), Royal University of Agriculture (RUA)).			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)? >>	**No**	No, biodigesters are only built at the premise of a household farm and animal farms.	N/A

Expert stakeholder Opinion for Principle 4.2: - No risks, the biodigesters are constructed and installed in the garden of the farmer, taking a plot as small as 5 by 4 meters for domestic biodigester and for medium and large scale biodigester, they are built within the premises of existing animal farms (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia). - Agreed to the above statement, the implementation of biodigesters confined to household farms and animal farms, does not necessitate or result in the physical or economic relocation of any individuals, either temporality, in any capacity (opinion from Mr. Lytour Lor, director, BTIC, RUA).			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	**No**	No, as above, the project does not result in a change in land tenure rights	N/A
Expert stakeholder Opinion for Principle 4.3: - No risks, the biodigesters are constructed and installed in the garden of the farmer, taking a plot as small as 5 by 4 meters for domestic biodigester and for medium and large scale biodigester, they are built within the premises of existing animal farms. There is no risk on land tenure rights (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia). - Agreed to the above statement, as previously mentioned, the Project's implementation, centered around biodigesters situated on household farms and animal farms, ensures that there are no requirements for changes in land tenure arrangements, access rights, usage rights, or land ownership (opinion from Mr. Lytour Lor, director, BTIC, RUA).			

Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	**Yes**	Project are located individually each households back yard or animal farms which are owned by indigenous people and also in household level where they also live is the same place.	No need to monitor, because project is implemented directly indigenous peoples living area or their farms by purchasing the project technology with their own consent and based on voluntary activity .

Expert stakeholder Opinion for Principle 4.4:

- No risks, the biodigesters are constructed and installed in the garden of the farmer, taking a plot as small as 5 by 4 meters for domestic biodigester and for medium and large scale biodigester, they are built within the premises of existing animal farms. There is no risk for indigenous people (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia).
- Agreed to the above statement, The projects are situated individually in each household's backyard or on animal farms owned by indigenous people, often within the same area where they reside. Domestic-scale biodigester present no discernible risk to indigenous communities in these locations (opinion from Mr. Lytour Lor, director, BTIC, RUA).

Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	Cambodia ratified the United Nations Convention against Corruption in September 2007[39] The NBP, PPBO, BCA masons etc. do not engage in any type of corruption activities	N/A
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards	No	1. Yes, working conditions of project employees and masons are in compliance with national labour law and occupational health and safety laws. 2. Yes, they are allowed to join any organisations, union or labour. Cambodia ratified the ILO Convention C087 (Freedom of Association and Protection of the Right to Organise Convention, 1948) and C098 (Right to Organise and Collective Bargaining Convention, 1949)[40] 3.BCAs and masons are independent from NBP, however the franchise contract with NBP stipulates that they have to follow the labour law. There is no provision for overtime – as workers are all freelance. An accident insurance is provided to the masons by the BCA. No, but masons work on a freelance basis, and work is often one digester or a few at the time 4. Yes, all is the employment model is culturally appropriate	N/A

[39] http://www.unodc.org/unodc/en/corruption/ratification-status.html

[40] http://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:11200:0::NO::P11200_COUNTRY_ID:103055

embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND		5. Certification of masons is only possible aged 18 and above. The certification is the ID card. NBP verifies this with the ID card or birth certificate.	

d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall			

ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation? >>	No	1. Yes, warranty is still offered after the crediting period by locking a after warranty fee for BCAs that will only be released after the warranty period. Furthermore, once a biodigester is constructed, they last for around 20 years. 2. No risks are attributed or associated with this project. On the contrary skilled and non-skilled employment is created benefitting the rural economics reducing the need to migrate for good quality jobs.	N/A
Principle 7.1 Emissions			
Will the Project increase greenhouse gas	No	The project decrease the GHG emissions, where it all calculated above in three categories of activities; Thermal,	N/A

emissions over the Baseline Scenario?		AWMS and electrical applications are implemented to create emission reductions.	
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	Project is not connected to local grid or use energy from fuel source that provides for other local users.	N/A
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	1. No the construction of biodigesters does not affect natural or pre-existing pattern of watercourses, ground-water and/or the watershed	N/A

>>			
Expert Stakeholder Opinion for Principle 8.1: - The biodigesters have not risk on the water flows or on the water shed. In Fact, pollutants are reduced in the process (BOD, COD, pathogens), (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia). - Agreed. No risks, the construction of biodigesters does not disrupt the natural or pre-existing patterns of water courses, groundwater, or the watershed (opinion from Mr. Lytour Lor, director, BTIC, RUA).			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	No, the construction occurs in backyards of households and next to the animal farms, and does not cause erosion	N/A
>>			
Expert Stakeholder Opinion for Principle 8.2: - There are no erosion risks related to biodigesters (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia). - Agreed. There are no risks associated with the construction of biodigesters in the backyards of households or adjacent to animal farms, as it does not contribute to erosion- related concerns (opinion from Mr. Lytour Lor, director, BTIC, RUA).			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil	No	No, the construction of biodigesters does not use land and soil or crops.	N/A

for production of crops or other products? >>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions? >>	No	No, the construction of biodigesters is not related to these risks.	N/A
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include	No	This is not applicable to the project as it does not produce crops and does not involve any use of GMOs.	N/A

GMOs in their processes and production)?			
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	no	Digester effluent can cause local eutrophication if not use similar to the baseline situation. However, most farmers use bio-slurry which is a superior fertilizer compared to farmyard manure and the impact is therefore less on the environment compared to the baseline. In the animal farms, they operate with a license where they follow veterinary technical standards. This standard covers the good practice of AWMS and they do not release any pollutants.	N/A
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	No hazardous and non-hazardous chemical are involved in biodigester use and construction	N/A
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	Farmers are trained to use bio-slurry as effective organic fertilizer which can improve yields and soil quality. However, the project does not encourage using chemical fertilizers.	N/A
>>			

Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	On the contrary, the project will result in a lower demand for firewood.	
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	On the contrary, bio-slurry is a very good fertilizer which improves crop quality	N/A
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project involves the construction of biodigesters through the development of a private sector. Households that included in the project sometimes raise animals, but the project only focusses on the manure excreted and managed	N/A
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems,	No	Not applicable, the project only built digesters in backyards of household farmers and vicinity in the animal farms, generally next to animal barns to attain direct flow.	N/A

critical habitats, landscapes, key biodiversity areas or sites identified?			
>>			

Expert Stakeholder Opinion for Principle 9.10:

- Habitats are not impacted by biodigesters, there is no such risk (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia).
- Agreed. The project implementation focus on constructing digesters in the backyards of household farmers and in close proximity to animal farms, especially adjacent to animal barns for direct flow, does not impact largely intact or hight conservation Value (HLV) ecosystems, critical habitats, landscapes, key biodiversity areas, or identified sites (opinion from Mr. Lytour Lor, director, BTIC, RUA).

Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other	No	No, the project only focusses on rural areas with technical potential for biogas. These exclude area where endangered species may live	N/A

areas where endangered species may be present through transboundary affects?			
>>			
Expert Stakeholder Opinion for Principle 9.11: - Biodigesters don’t endanger species, on the contrary, replacing firewood and charcoal with biogas removes the need to forage in the forests, the place where endangered species could live (opinion from Mr. Bastiaan Tueune, Sector Leader Energy at SNV in Cambodia). - Agreed. No endangered species have been identified as potentially present within the Project boundary, including those that may traverse the area. Further, the Project’s focus solely on rural aeras with technical potential for biogas excluded any impact on areas where endangered species may reside, thus avoiding transboundary effects on their habitats. (opinion from Mr. Lytour Lor, director, BTIC, RUA).			

APPENDIX 2 - CONTACT INFORMATION OF PROJECT DEVELOPER(S)

Organization name	National Biodigester Programme
Registration number with relevant authority	K007-100070698
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APPENDIX 3 - LUF ADDITIONAL INFORMATION

N/A

APPENDIX 4 - DESIGN CHANGES

N/A

Revision History

Version	Date	Remarks
1.3	14 April 2023	Integrated the design change memo as annex of the document. Editorial changes
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption